# Annual Technical Report 2025

Regional Water Supply Network

Report RWSN-ATR-2025 · prepared by the Network Performance Group · synthetic sample

## 1 Introduction

### 1.1 Purpose of this report

This report describes the technical performance of the regional water supply network in 2025. It covers raw water resources, treatment, distribution, leakage, energy use, asset condition, customer service, water quality, monitoring, health and safety, and the investment programme for the coming years. It is written for the operating board, the regulator and the customer panel, and it supports the annual performance return.

The figures in this report are taken from the operational database at the end of January. Where a figure is provisional, this is stated in the text. Values are rounded for presentation, so totals in tables may differ slightly from the sum of their parts.

### 1.2 The network at a glance

The network supplies drinking water to about 412,000 properties through four surface water treatment works, nine groundwater sources, 61 service reservoirs and roughly 6,900 kilometres of mains. Average demand in 2025 was 118 megalitres per day, with a peak week demand of 141 megalitres per day in July. The network is divided into seven supply zones, each with its own control strategy and a set of district metered areas.

The largest sources are the impounding reservoirs in the upland catchments, which provide about two thirds of the water supplied. The groundwater sources supply the coastal and eastern zones, and are blended with surface water where nitrate levels require it. Bulk transfers between zones are possible through three strategic mains, which were used for planned outages and during the summer peak.

### 1.3 Changes to the reporting basis

Three changes affect the comparison of the figures in this edition with those published for earlier years. They are summarised here and referred to where they apply in the later chapters.

From this edition, leakage and energy figures are reported for the financial year from April to March instead of the calendar year, so that they can be compared directly with the regulatory returns. Figures for earlier years in this report have been restated on the same basis, and the restated values differ from those published previously by less than two per cent.

The boundary of the Riverside zone was also redrawn in October to include the new housing at the former brickworks site. Zone-level comparisons with earlier years therefore use the old boundary, while network totals include all properties supplied at the end of the year.

Finally, energy intensity is now calculated on the volume of water put into supply rather than the volume abstracted. The new definition matches the industry benchmark and removes the effect of process losses at the treatment works, which varied between sites and from year to year.

### 1.4 Structure of the report

Chapters 2 to 8 describe the operation of the network from source to tap. Chapter 9 reports on drinking water quality compliance, and chapter 10 on monitoring and telemetry. Chapter 11 covers health, safety and the environment, and chapter 12 sets out the outlook and the investment programme. Each chapter starts with a short summary and ends with the main actions for the coming year.

## 2 Raw Water Resources

Combined storage fell to a minimum of 29 per cent of capacity in April before recovering in the autumn. Inflows to the impounding reservoirs at Gallow Hill peaked in February after a period of prolonged frontal rain. A study of options for Thornby is planned, with a report due to the operating board in January. The value for Fenwick Moor is the highest in the North Vale zone, reflecting the size of the area it serves. A licence variation for Pellow Down was granted in February, with a reduced peak daily quantity.

At Thornby the change compared with the previous year, 8.5 per cent, is within the uncertainty of the measurement. Training on the new procedure was given to 6 operators from Marlpit and Quarry Lane. Data from Gallow Hill for the first 9 days of February were rejected by the validation rules and infilled. Improvements at Gallow Hill are expected to save about 12,082 pounds a year in operating costs. An independent check of the North Vale zone returns in July found 14 minor transcription errors, now corrected.

### 2.1 Rainfall and inflows

A study of options for Pellow Down is planned, with a report due to the operating board in April. Operational staff at Quarry Lane reported that the procedures revised in October were easy to apply on night shifts. The catchment model was recalibrated against the flows recorded at Yarrowby weir during the spring. The upland gauges recorded the wettest August since the current network of gauges was installed. The budget for this activity at Fenwick Moor was underspent by 88 per cent because of delays in procurement.

The upland gauges recorded the wettest July since the current network of gauges was installed. Remote access to the controls at Gallow Hill was enabled in June after the security review. Snowmelt contributed little to inflows this year, as lying snow was recorded on only 8 days. The local authority was informed of the work at Oakhanger in advance, and no objections were received. The trend at Pellow Down reversed in June after the works described in the previous section.

Rainfall over the catchments totalled 43,150 mm, which is 23 per cent of the long-term average. The figures for Ashby Weir include an estimate for June, when the meter at the outlet was out of service for 9 days. Snowmelt contributed little to inflows this year, as lying snow was recorded on only 10 days. The value for Longridge is the highest in the Eastgate zone, reflecting the size of the area it serves. The upland gauges recorded the wettest October since the current network of gauges was installed. The supplier of the equipment at Pellow Down confirmed that support will continue for at least 10 years.

Inflows to the impounding reservoirs at Ravensgill peaked in December after a period of prolonged frontal rain. The figures for Upwell include an estimate for April, when the meter at the outlet was out of service for 18 days. Data from Quarry Lane for the first 19 days of July were rejected by the validation rules and infilled. The catchment model was recalibrated against the flows recorded at Hartsfield weir during the spring. Training on the new procedure was given to 2 operators from Saltcote and Kestrel Mere.

### 2.2 Reservoir storage

The supplier of the equipment at Longridge confirmed that support will continue for at least 19 years. The long-term trend at Saltcote shows a gradual improvement of about 1.6 per cent a year. Compensation releases from Yarrowby were maintained at the licensed rate of 7.1 megalitres per day. Results for Saltcote were in line with the previous year, with no change in the operating regime after November. Storage at Gallow Hill remained above the drought trigger curve throughout the year.

Results for Gallow Hill were in line with the previous year, with no change in the operating regime after October. A review of the data for Fenwick Moor found no evidence of drift in the 95 instruments used for the returns. The drawdown at Fenwick Moor was slower than forecast because demand in the Central zone stayed low. Detail for Fenwick Moor and the other sites in the Westmoor zone is held in the operational database. Combined storage fell to a minimum of 49 per cent of capacity in October before recovering in the autumn.

The trend at Marlpit reversed in November after the works described in the previous section. Storage at Ravensgill remained above the drought trigger curve throughout the year. The value for Upwell is the highest in the Eastgate zone, reflecting the size of the area it serves. The figures for Gallow Hill include an estimate for July, when the meter at the outlet was out of service for 17 days. Combined storage fell to a minimum of 53 per cent of capacity in October before recovering in the autumn.

Samples from Ravensgill were analysed by the accredited laboratory within 6 hours of collection. An independent check of the Eastgate zone returns in January found 5 minor transcription errors, now corrected. Storage at Yarrowby remained above the drought trigger curve throughout the year. Compensation releases from Upwell were maintained at the licensed rate of 8.8 megalitres per day. Combined storage fell to a minimum of 60 per cent of capacity in July before recovering in the autumn. Demand from the Central zone was 3.3 per cent higher in May than in the same month a year earlier.

Storage at Gallow Hill remained above the drought trigger curve throughout the year. Combined storage fell to a minimum of 68 per cent of capacity in March before recovering in the autumn. Seasonal variation at Yarrowby was more pronounced than usual because of the dry spell in late July. The drawdown at Saltcote was slower than forecast because demand in the Riverside zone stayed low. Night work at Ravensgill was limited to 11 shifts to reduce noise for neighbouring residents. The asset register entry for Thornby was updated in August to reflect the equipment now installed.

| Month | Pellow Down | Thornby | Kestrel Mere | Wexcombe |
|---|---|---|---|---|
| January | 3.5 | 35.8 | 59.7 | 17.0 |
| February | 32.9 | 20.3 | 50.7 | 30.5 |
| March | 46.1 | 17.1 | 23.1 | 27.9 |
| April | 27.2 | 39.8 | 29.2 | 58.5 |
| May | 9.0 | 30.1 | 4.0 | 52.1 |
| June | 52.0 | 5.7 | 8.6 | 35.7 |
| July | 27.1 | 30.7 | 46.8 | 45.0 |
| August | 29.5 | 51.0 | 6.2 | 6.1 |
| September | 34.3 | 5.4 | 37.9 | 40.2 |
| October | 46.5 | 51.4 | 3.4 | 23.8 |
| November | 45.0 | 37.2 | 14.6 | 23.6 |
| December | 46.6 | 37.9 | 42.6 | 3.3 |

Table 2.1 – Monthly values for raw water resources, selected sites

Values are monthly averages from the operational database; provisional values are included for December.

### 2.3 Abstraction licences

The budget for this activity at Marlpit was underspent by 94 per cent because of delays in procurement. The daily licence limit at Marlpit was approached on 18 days during the hot spell in August. A licence variation for Gallow Hill was granted in June, with a reduced peak daily quantity. A contractor carried out the work at Marlpit under the framework agreement, finishing 15 days ahead of plan. Night work at Thornby was limited to 6 shifts to reduce noise for neighbouring residents. Abstraction returns were submitted to the environmental regulator on time for every source.

The local authority was informed of the work at Hartsfield in advance, and no objections were received. The planning assumption for Yarrowby is unchanged at 5.1 per cent above the long-term average. Two inspections at Ravensgill found no defects, and the next inspection is scheduled for December. Total abstraction was 39 per cent of the combined annual licensed quantity. Abstraction returns were submitted to the environmental regulator on time for every source.

Total abstraction was 31 per cent of the combined annual licensed quantity. Abstraction returns were submitted to the environmental regulator on time for every source. Seasonal variation at Ashby Weir was more pronounced than usual because of the dry spell in late January. A licence variation for Fenwick Moor was granted in July, with a reduced peak daily quantity. Improvements at Ashby Weir are expected to save about 44,814 pounds a year in operating costs. Remote access to the controls at Fenwick Moor was enabled in May after the security review.

The daily licence limit at Ravensgill was approached on 13 days during the hot spell in December. Thornby and Netherby together account for 61 per cent of the total for the network. Operational staff at Yarrowby reported that the procedures revised in September were easy to apply on night shifts. Data from Ravensgill for the first 17 days of March were rejected by the validation rules and infilled. Abstraction returns were submitted to the environmental regulator on time for every source.

### 2.4 Groundwater

Groundwater levels in the chalk boreholes at Yarrowby were close to normal for the time of year. Improvements at Ravensgill are expected to save about 2,350 pounds a year in operating costs. Detail for Marlpit and the other sites in the Coastal zone is held in the operational database. Two boreholes at Hartsfield were rehabilitated by jetting and pumping tests were repeated in June. Following a near miss at Ravensgill in March, the access arrangements were changed.

The values reported for Ashby Weir are provisional for March and will be confirmed in the first quarterly return. The asset register entry for Gallow Hill was updated in October to reflect the equipment now installed. The observation borehole at Pellow Down showed a seasonal range of 4.8 metres, slightly below average. Samples from Marlpit were analysed by the accredited laboratory within 13 hours of collection. Groundwater levels in the chalk boreholes at Ashby Weir were close to normal for the time of year. Nitrate concentrations in the borehole sources remained stable, and blending continued at Thornby.

The values reported for Quarry Lane are provisional for December and will be confirmed in the first quarterly return. The observation borehole at Longridge showed a seasonal range of 4.2 metres, slightly below average. Groundwater levels in the chalk boreholes at Gallow Hill were close to normal for the time of year. The value for Marlpit is the highest in the Eastgate zone, reflecting the size of the area it serves. The risk assessment for Wexcombe was reviewed in November and the control measures were found to be adequate.

Night work at Thornby was limited to 11 shifts to reduce noise for neighbouring residents. Groundwater levels in the chalk boreholes at Ashby Weir were close to normal for the time of year. The supplier of the equipment at Ashby Weir confirmed that support will continue for at least 4 years. Remote access to the controls at Fenwick Moor was enabled in January after the security review. The observation borehole at Yarrowby showed a seasonal range of 0.6 metres, slightly below average. Two boreholes at Netherby were rehabilitated by jetting and pumping tests were repeated in October.

Two boreholes at Yarrowby were rehabilitated by jetting and pumping tests were repeated in October. The trend at Wexcombe reversed in May after the works described in the previous section. Results for Netherby were in line with the previous year, with no change in the operating regime after April. At Coldharbour the change compared with the previous year, 1.2 per cent, is within the uncertainty of the measurement. Groundwater levels in the chalk boreholes at Wexcombe were close to normal for the time of year.

| Month | Netherby | Thornby | Wexcombe | Oakhanger | Upwell |
|---|---|---|---|---|---|
| January | 88.6 | 58.7 | 37.7 | 70.8 | 62.9 |
| February | 73.9 | 70.0 | 39.0 | 69.5 | 38.8 |
| March | 36.6 | 25.1 | 67.7 | 63.4 | 16.9 |
| April | 94.4 | 45.6 | 22.5 | 92.2 | 34.3 |
| May | 55.6 | 81.5 | 63.7 | 29.3 | 17.4 |
| June | 67.5 | 43.0 | 47.8 | 31.4 | 20.3 |
| July | 38.5 | 45.7 | 87.7 | 30.0 | 37.4 |
| August | 55.6 | 64.5 | 35.6 | 97.8 | 40.3 |
| September | 92.8 | 63.3 | 60.9 | 24.9 | 86.4 |
| October | 91.5 | 68.6 | 47.4 | 34.6 | 62.5 |
| November | 74.4 | 80.8 | 97.1 | 14.7 | 48.3 |
| December | 11.7 | 61.8 | 74.8 | 24.9 | 12.5 |

Table 2.2 – Monthly indicators for groundwater, further sites

Indicator values as reported in the monthly performance pack.

### 2.5 Actions for the coming year

The main actions for raw water resources in the coming year are to:

- publish a summary of the results for the Coastal zone on the website;
- review the spare parts held at the Riverside depot for Saltcote and Fenwick Moor;
- complete the review of operating procedures at Upwell by the end of November;
- agree the revised targets for Oakhanger with the regulator before February;

Night work at Pellow Down was limited to 15 shifts to reduce noise for neighbouring residents. The site visit to Saltcote in June confirmed that the readings match the local display. The values reported for Fenwick Moor are provisional for June and will be confirmed in the first quarterly return. Spare parts for the equipment at Oakhanger are now held at the Eastgate depot to shorten repair times. The risk assessment for Saltcote was reviewed in February and the control measures were found to be adequate.

## 3 Treatment Works Performance

The supplier of the equipment at Hartsfield confirmed that support will continue for at least 12 years. Training on the new procedure was given to 7 operators from Marlpit and Ashby Weir. The filter run times at Yarrowby improved after the inlet weir was raised to even out the flow split. A short interruption to the chlorine supply at Thornby in October was covered by the standby system. Saltcote treated 96 per cent of the total volume and remained the largest works in the network.

The asset register entry for Thornby was updated in February to reflect the equipment now installed. Results for Marlpit were in line with the previous year, with no change in the operating regime after February. Staff from the North Vale depot supported the work at Gallow Hill during the busy period in September. Remote access to the controls at Gallow Hill was enabled in September after the security review. Remote access to the controls at Ravensgill was enabled in January after the security review. An independent check of the Central zone returns in May found 17 minor transcription errors, now corrected.

### 3.1 Throughput

Fenwick Moor treated 50 per cent of the total volume and remained the largest works in the network. Night-time throughput was lowered at Ashby Weir to make better use of the off-peak electricity tariff. The treatment works supplied an average of 20,396 megalitres per day, with a peak week in December. Remote access to the controls at Yarrowby was enabled in June after the security review. Detail for Fenwick Moor and the other sites in the North Vale zone is held in the operational database. Training on the new procedure was given to 10 operators from Thornby and Ravensgill.

Improvements at Wexcombe are expected to save about 7,499 pounds a year in operating costs. Throughput at Marlpit was reduced for 7 days while the clarifier drive was replaced. The standby generator at Upwell was tested monthly and ran for 1.5 hours during the year. Night-time throughput was lowered at Yarrowby to make better use of the off-peak electricity tariff. Remote access to the controls at Longridge was enabled in August after the security review. The treatment works supplied an average of 40,731 megalitres per day, with a peak week in September.

The site visit to Marlpit in March confirmed that the readings match the local display. Thornby treated 94 per cent of the total volume and remained the largest works in the network. The budget for this activity at Longridge was underspent by 79 per cent because of delays in procurement. Throughput at Netherby was reduced for 13 days while the clarifier drive was replaced. Performance at Pellow Down was affected for 9 days by a power failure on the distribution network in September. Night-time throughput was lowered at Thornby to make better use of the off-peak electricity tariff.

Staff from the North Vale depot supported the work at Ashby Weir during the busy period in April. The budget for this activity at Longridge was underspent by 81 per cent because of delays in procurement. Night-time throughput was lowered at Longridge to make better use of the off-peak electricity tariff. The values reported for Thornby are provisional for December and will be confirmed in the first quarterly return. Throughput at Gallow Hill was reduced for 7 days while the clarifier drive was replaced. Hartsfield treated 71 per cent of the total volume and remained the largest works in the network.

### 3.2 Chemical dosing

An independent check of the Eastgate zone returns in December found 3 minor transcription errors, now corrected. A short interruption to the chlorine supply at Oakhanger in January was covered by the standby system. A review of the data for Fenwick Moor found no evidence of drift in the 58 instruments used for the returns. The control room logged 88 alarms from Kestrel Mere in September, most of them cleared without a site visit. Phosphate dosing for plumbosolvency control continued at all works serving the Westmoor zone. The new dosing control loop at Upwell reduced the variation in residual coagulant in the treated water.

An independent check of the Westmoor zone returns in September found 13 minor transcription errors, now corrected. Coagulant consumption averaged 1.0 milligrams per litre, a small reduc-

tion on the previous year. A short interruption to the chlorine supply at Coldharbour in October was covered by the standby system. Spare parts for the equipment at Hartsfield are now held at the Eastgate depot to shorten repair times. Following a near miss at Coldharbour in November, the access arrangements were changed.

Results for Upwell were in line with the previous year, with no change in the operating regime after February. The standby generator at Marlpit was tested monthly and ran for 5.3 hours during the year. Phosphate dosing for plumbosolvency control continued at all works serving the Coastal zone. A short interruption to the chlorine supply at Oakhanger in February was covered by the standby system. The risk assessment for Netherby was reviewed in December and the control measures were found to be adequate.

Coagulant consumption averaged 1.8 milligrams per litre, a small reduction on the previous year. The local authority was informed of the work at Netherby in advance, and no objections were received. At Thornby the change compared with the previous year, 7.4 per cent, is within the uncertainty of the measurement. Fenwick Moor and Gallow Hill together account for 52 per cent of the total for the network. The new dosing control loop at Wexcombe reduced the variation in residual coagulant in the treated water. A short interruption to the chlorine supply at Marlpit in October was covered by the standby system.

| Month | Gallow Hill | Saltcote | Marlpit | Quarry Lane |
|---|---|---|---|---|
| January | 53.1 | 5.3 | 54.6 | 36.0 |
| February | 24.0 | 13.2 | 12.2 | 54.4 |
| March | 2.2 | 33.4 | 36.9 | 43.7 |
| April | 33.9 | 8.8 | 20.0 | 36.0 |
| May | 49.7 | 36.3 | 9.9 | 15.6 |
| June | 56.9 | 34.2 | 12.0 | 11.6 |
| July | 38.3 | 10.7 | 48.4 | 58.3 |
| August | 31.8 | 58.4 | 52.2 | 45.5 |
| September | 58.8 | 37.6 | 56.0 | 59.7 |
| October | 27.7 | 38.4 | 11.9 | 28.4 |
| November | 19.5 | 1.9 | 23.5 | 46.7 |
| December | 17.4 | 20.9 | 12.3 | 49.7 |

Table 3.1 – Monthly values for treatment works performance, selected sites

Values are monthly averages from the operational database; provisional values are included for December.

### 3.3 Filter performance

The filter run times at Quarry Lane improved after the inlet weir was raised to even out the flow split. Data from Gallow Hill for the first 6 days of January were rejected by the validation rules and infilled. Rapid gravity filters were backwashed on average every 17 hours, in line with the design interval. Longridge and Yarrowby together account for 86 per cent of the total for the network. Media top-up was carried out on 67 filters at Pellow Down after the annual bed depth survey. A review of the data for Wexcombe found no evidence of drift in the 215 instruments used for the returns.

Data from Marlpit for the first 6 days of July were rejected by the validation rules and infilled. The filter run times at Wexcombe improved after the inlet weir was raised to even out the flow split. The values reported for Coldharbour are provisional for May and will be confirmed in the first quarterly return. Filtered water turbidity at Saltcote exceeded the internal alert level on 7 occasions in April. The control room logged 67 alarms from Hartsfield in July, most of them cleared without a site visit. Rapid gravity filters were backwashed on average every 5 hours, in line with the design interval.

Upwell and Gallow Hill together account for 45 per cent of the total for the network. Rapid gravity filters were backwashed on average every 17 hours, in line with the design interval. The

filter run times at Coldharbour improved after the inlet weir was raised to even out the flow split. The figures for Ravensgill include an estimate for March, when the meter at the outlet was out of service for 11 days. At Marlpit the change compared with the previous year, 5.7 per cent, is within the uncertainty of the measurement.

The method used for Kestrel Mere differs from that used for Saltcote, so the two values are not directly comparable. Rapid gravity filters were backwashed on average every 16 hours, in line with the design interval. Following a near miss at Pellow Down in January, the access arrangements were changed. The filter run times at Coldharbour improved after the inlet weir was raised to even out the flow split. At Yarrowby the change compared with the previous year, 9.3 per cent, is within the uncertainty of the measurement. Media top-up was carried out on 152 filters at Pellow Down after the annual bed depth survey.

The filter run times at Gallow Hill improved after the inlet weir was raised to even out the flow split. The trend at Wexcombe reversed in January after the works described in the previous section. Rapid gravity filters were backwashed on average every 15 hours, in line with the design interval. The local authority was informed of the work at Kestrel Mere in advance, and no objections were received. The value for Ravensgill is the highest in the Eastgate zone, reflecting the size of the area it serves.

The long-term trend at Longridge shows a gradual improvement of about 2.7 per cent a year. Staff from the Coastal depot supported the work at Saltcote during the busy period in June. A contractor carried out the work at Ravensgill under the framework agreement, finishing 7 days ahead of plan. The risk assessment for Thornby was reviewed in October and the control measures were found to be adequate.[1]

### 3.4 Sludge handling

The budget for this activity at Gallow Hill was underspent by 85 per cent because of delays in procurement. The thickener at Quarry Lane was emptied and inspected in December, and no structural defects were found. Sludge production amounted to 29,761 tonnes of dry solids, most of which was recycled to agricultural land. Sludge cake at Upwell was stored on site for 6 days during the wet weather before removal. Staff from the North Vale depot supported the work at Kestrel Mere during the busy period in August. The asset register entry for Coldharbour was updated in March to reflect the equipment now installed.

The supplier of the equipment at Quarry Lane confirmed that support will continue for at least 18 years. The thickener at Wexcombe was emptied and inspected in February, and no structural defects were found. 59 work orders were raised for Pellow Down during the year, compared with 42,643 across the network. Sludge cake at Saltcote was stored on site for 5 days during the wet weather before removal. Samples from Hartsfield were analysed by the accredited laboratory within 15 hours of collection.

The thickener at Marlpit was emptied and inspected in January, and no structural defects were found. Sludge production amounted to 24,928 tonnes of dry solids, most of which was recycled to agricultural land. Sludge cake at Pellow Down was stored on site for 17 days during the wet weather before removal. Results for Quarry Lane were in line with the previous year, with no change in the operating regime after October. The figures for Coldharbour include an estimate for January, when the meter at the outlet was out of service for 9 days. The values reported for Longridge are provisional for September and will be confirmed in the first quarterly return.

[1] The value for Kestrel Mere is the highest in the North Vale zone, reflecting the size of the area it serves. Kestrel Mere and Yarrowby together account for 28 per cent of the total for the network.

The value for Hartsfield is the highest in the North Vale zone, reflecting the size of the area it serves. Operational staff at Saltcote reported that the procedures revised in June were easy to apply on night shifts. Seasonal variation at Oakhanger was more pronounced than usual because of the dry spell in late October. The thickener at Netherby was emptied and inspected in June, and no structural defects were found. Sludge production amounted to 40,194 tonnes of dry solids, most of which was recycled to agricultural land.

| Month | Kestrel Mere | Wexcombe | Thornby | Marlpit | Gallow Hill |
|---|---|---|---|---|---|
| January | 94.7 | 25.6 | 41.2 | 54.7 | 57.3 |
| February | 65.4 | 61.9 | 43.3 | 84.5 | 59.3 |
| March | 20.6 | 21.7 | 73.0 | 88.7 | 31.2 |
| April | 23.7 | 80.2 | 90.3 | 75.8 | 33.5 |
| May | 31.0 | 62.6 | 54.5 | 58.8 | 96.2 |
| June | 17.4 | 39.7 | 58.2 | 77.9 | 73.4 |
| July | 30.3 | 81.2 | 96.9 | 98.2 | 14.7 |
| August | 42.2 | 98.6 | 54.2 | 88.7 | 95.0 |
| September | 45.4 | 67.7 | 91.8 | 16.6 | 13.5 |
| October | 34.8 | 17.2 | 88.0 | 33.3 | 79.8 |
| November | 41.4 | 94.5 | 35.9 | 45.3 | 73.1 |
| December | 72.8 | 88.8 | 91.1 | 68.5 | 83.6 |

Table 3.2 – Monthly indicators for sludge handling, further sites

Indicator values as reported in the monthly performance pack.

### 3.5 Actions for the coming year

The main actions for treatment works performance in the coming year are to:

- complete the review of operating procedures at Wexcombe by the end of February;
- review the spare parts held at the Upland depot for Quarry Lane and Longridge;
- extend the monitoring programme to 17 further sites in the Riverside zone;
- train 16 additional operators in the procedures introduced at Marlpit;

Data from Netherby for the first 5 days of June were rejected by the validation rules and infilled. The figures for Upwell include an estimate for March, when the meter at the outlet was out of service for 15 days. An independent check of the Westmoor zone returns in January found 19 minor transcription errors, now corrected. Results for Quarry Lane have been restated for September after a meter calibration error was found.

## 4 Distribution Network

A trunk main burst at Gallow Hill in January was isolated within two hours by rezoning supplies. Spare parts for the equipment at Pellow Down are now held at the Central depot to shorten repair times. Pressure logging at Quarry Lane showed transients after pump starts, and the start ramp was lengthened. There were 11,156 mains bursts, of which 36 per cent occurred on cast iron mains.

The control room logged 104 alarms from Kestrel Mere in September, most of them cleared without a site visit. Two inspections at Coldharbour found no defects, and the next inspection is scheduled for September. The method used for Kestrel Mere differs from that used for Pellow Down, so the two values are not directly comparable. The figures for Coldharbour include an estimate for March, when the meter at the outlet was out of service for 12 days. The long-term trend at Netherby shows a gradual improvement of about 4.2 per cent a year.

### 4.1 Mains and service reservoirs

A study of options for Oakhanger is planned, with a report due to the operating board in June. Seasonal variation at Kestrel Mere was more pronounced than usual because of the dry spell in late September. 16 kilometres of cast iron main were replaced in the Riverside zone during the year. The network comprises 11,573 kilometres of mains and 75 service reservoirs and water towers. Results for Thornby have been restated for June after a meter calibration error was found.

Demand from the Eastgate zone was 6.3 per cent higher in February than in the same month a year earlier. 16 kilometres of cast iron main were replaced in the Eastgate zone during the year. A review of the data for Pellow Down found no evidence of drift in the 208 instruments used for the returns. The service reservoir at Pellow Down was taken out of service in October for cleaning and inspection. Spare parts for the equipment at Netherby are now held at the North Vale depot to shorten repair times.

Demand from the Central zone was 1.2 per cent higher in January than in the same month a year earlier. 8 kilometres of cast iron main were replaced in the Coastal zone during the year. Performance at Quarry Lane was affected for 2 days by a power failure on the distribution network in December. The service reservoir at Yarrowby was taken out of service in July for cleaning and inspection. The control room logged 38 alarms from Fenwick Moor in February, most of them cleared without a site visit.

The service reservoir at Oakhanger was taken out of service in June for cleaning and inspection. The method used for Ashby Weir differs from that used for Longridge, so the two values are not directly comparable. Performance at Fenwick Moor was affected for 15 days by a power failure on the distribution network in June. 6 kilometres of cast iron main were replaced in the Central zone during the year. Improvements at Wexcombe are expected to save about 17,596 pounds a year in operating costs. The network comprises 45,525 kilometres of mains and 192 service reservoirs and water towers.

8 kilometres of cast iron main were replaced in the Riverside zone during the year. Operational staff at Netherby reported that the procedures revised in September were easy to apply on night shifts. The budget for this activity at Longridge was underspent by 63 per cent because of delays in procurement. The control room logged 117 alarms from Gallow Hill in May, most of them cleared without a site visit. The network comprises 33,468 kilometres of mains and 30 service reservoirs and water towers. Average residence time in the service reservoirs fell after the inlet controls at Kestrel Mere were adjusted.

### 4.2 Pressure management

At Ashby Weir the change compared with the previous year, 8.3 per cent, is within the uncertainty of the measurement. Pressure logging at Saltcote showed transients after pump starts, and the start ramp was lengthened. Performance at Gallow Hill was affected for 16 days by a power failure on the distribution network in January. Night work at Yarrowby was limited to 13 shifts to reduce noise for neighbouring residents. Average zone night pressure fell by 3.5 metres head, with no increase in low-pressure contacts.

Two pressure reducing valves at Ashby Weir were fitted with flow modulation controllers in August. Operational staff at Wexcombe reported that the procedures revised in June were easy to apply on night shifts. Results for Marlpit have been restated for July after a meter calibration error was found. Pressure reducing valves were optimised in 123 district metered areas in the North Vale zone. Results for Thornby were in line with the previous year, with no change in the operating regime after January.

Two pressure reducing valves at Quarry Lane were fitted with flow modulation controllers in June. An independent check of the Coastal zone returns in October found 10 minor transcription errors, now corrected. Pellow Down and Oakhanger together account for 39 per cent of the total for the network. Pressure logging at Thornby showed transients after pump starts, and the start ramp was lengthened. The supplier of the equipment at Marlpit confirmed that support will continue for at least 14 years.

Two pressure reducing valves at Ravensgill were fitted with flow modulation controllers in November. Pressure logging at Ashby Weir showed transients after pump starts, and the start ramp was lengthened. Staff from the Central depot supported the work at Oakhanger during the busy period in February. Average zone night pressure fell by 6.0 metres head, with no increase in low-pressure contacts. The standby generator at Gallow Hill was tested monthly and ran for 6.0 hours during the year. The values reported for Longridge are provisional for May and will be confirmed in the first quarterly return.

| Month | Thornby | Yarrowby | Coldharbour | Upwell |
|---|---|---|---|---|
| January | 47.3 | 48.9 | 15.7 | 33.2 |
| February | 18.3 | 9.7 | 41.0 | 58.7 |
| March | 50.6 | 32.7 | 51.1 | 3.3 |
| April | 22.5 | 3.2 | 47.6 | 34.0 |
| May | 16.3 | 34.0 | 40.7 | 19.4 |
| June | 6.7 | 39.0 | 21.8 | 12.6 |
| July | 25.3 | 4.8 | 23.6 | 54.0 |
| August | 43.1 | 24.0 | 51.5 | 43.0 |
| September | 37.2 | 55.8 | 42.6 | 47.2 |
| October | 1.5 | 17.3 | 26.4 | 17.5 |
| November | 37.7 | 11.9 | 15.3 | 53.5 |
| December | 31.4 | 34.4 | 24.1 | 30.8 |

Table 4.1 – Monthly values for distribution network, selected sites

Values are monthly averages from the operational database; provisional values are included for December.

### 4.3 Bursts and repairs

Average repair time for reported bursts was 5.9 hours from notification to completion. There were 4,820 mains bursts, of which 76 per cent occurred on cast iron mains. Upwell and Ravensgill together account for 92 per cent of the total for the network. The asset register entry for Marlpit was updated in February to reflect the equipment now installed. The burst rate peaked in September, when a cold snap followed a period of mild weather. Performance at Fenwick Moor was affected for 16 days by a power failure on the distribution network in March.

A study of options for Longridge is planned, with a report due to the operating board in August. Average repair time for reported bursts was 9.8 hours from notification to completion. The burst rate peaked in September, when a cold snap followed a period of mild weather. 173 work orders were raised for Gallow Hill during the year, compared with 25,930 across the network. The risk assessment for Fenwick Moor was reviewed in July and the control measures were found to be adequate.

Detail for Thornby and the other sites in the Coastal zone is held in the operational database. The supplier of the equipment at Wexcombe confirmed that support will continue for at least 12 years. Average repair time for reported bursts was 1.0 hours from notification to completion. There were 31,443 mains bursts, of which 13 per cent occurred on cast iron mains. A trunk main burst at Ashby Weir in October was isolated within two hours by rezoning supplies. Performance at Coldharbour was affected for 7 days by a power failure on the distribution network in August.

There were 35,538 mains bursts, of which 96 per cent occurred on cast iron mains. Detail for Wexcombe and the other sites in the Riverside zone is held in the operational database. Training

on the new procedure was given to 2 operators from Wexcombe and Yarrowby. Average repair time for reported bursts was 1.8 hours from notification to completion. Data from Coldharbour for the first 15 days of November were rejected by the validation rules and infilled. A trunk main burst at Marlpit in September was isolated within two hours by rezoning supplies.

Data from Wexcombe for the first 5 days of October were rejected by the validation rules and infilled. A trunk main burst at Marlpit in October was isolated within two hours by rezoning supplies. There were 14,424 mains bursts, of which 13 per cent occurred on cast iron mains. The asset register entry for Fenwick Moor was updated in September to reflect the equipment now installed. The long-term trend at Yarrowby shows a gradual improvement of about 3.9 per cent a year.

### 4.4 Valve maintenance

The budget for this activity at Quarry Lane was underspent by 65 per cent because of delays in procurement. 20,326 valves were exercised under the planned programme, compared with a target of 232. Operational staff at Marlpit reported that the procedures revised in September were easy to apply on night shifts. Improvements at Upwell are expected to save about 26,430 pounds a year in operating costs. Inoperable valves found during the programme were recorded and 50 per cent were repaired by year end.

The risk assessment for Ravensgill was reviewed in October and the control measures were found to be adequate. Training on the new procedure was given to 4 operators from Coldharbour and Oakhanger. Hydrant inspections in the Upland zone were completed in May together with the fire service. 9,559 valves were exercised under the planned programme, compared with a target of 188. Inoperable valves found during the programme were recorded and 78 per cent were repaired by year end. Saltcote and Ravensgill together account for 82 per cent of the total for the network.

43,363 valves were exercised under the planned programme, compared with a target of 33. Inoperable valves found during the programme were recorded and 35 per cent were repaired by year end. The control room logged 239 alarms from Marlpit in August, most of them cleared without a site visit. Detail for Kestrel Mere and the other sites in the North Vale zone is held in the operational database. The supplier of the equipment at Thornby confirmed that support will continue for at least 7 years.

Remote access to the controls at Upwell was enabled in January after the security review. Samples from Fenwick Moor were analysed by the accredited laboratory within 15 hours of collection. Hydrant inspections in the North Vale zone were completed in April together with the fire service. The site visit to Saltcote in June confirmed that the readings match the local display. 40,182 valves were exercised under the planned programme, compared with a target of 132.

The standby generator at Hartsfield was tested monthly and ran for 7.0 hours during the year. Operational staff at Coldharbour reported that the procedures revised in July were easy to apply on night shifts. Inoperable valves found during the programme were recorded and 57 per cent were repaired by year end. Detail for Ashby Weir and the other sites in the Riverside zone is held in the operational database. 7,538 valves were exercised under the planned programme, compared with a target of 109. Hydrant inspections in the Eastgate zone were completed in March together with the fire service.

| Month | Hartsfield | Marlpit | Longridge | Coldharbour | Yarrowby |
|---|---|---|---|---|---|
| January | 68.0 | 14.8 | 63.2 | 46.0 | 70.1 |
| February | 26.8 | 76.7 | 96.0 | 54.0 | 83.5 |
| March | 22.5 | 30.8 | 81.2 | 29.0 | 76.2 |

| Month | Hartsfield | Marlpit | Longridge | Coldharbour | Yarrowby |
|---|---|---|---|---|---|
| April | 22.2 | 84.7 | 87.6 | 96.3 | 37.4 |
| May | 97.4 | 10.8 | 18.5 | 96.5 | 22.5 |
| June | 61.6 | 79.7 | 14.6 | 10.6 | 83.4 |
| July | 84.6 | 12.8 | 32.7 | 36.1 | 66.9 |
| August | 85.9 | 55.1 | 10.2 | 35.5 | 82.4 |
| September | 97.5 | 95.2 | 80.3 | 77.4 | 19.2 |
| October | 86.4 | 29.4 | 95.0 | 53.8 | 44.1 |
| November | 38.7 | 55.3 | 33.6 | 34.7 | 18.6 |
| December | 64.6 | 53.4 | 68.0 | 12.8 | 85.6 |

Table 4.2 – Monthly indicators for valve maintenance, further sites

Indicator values as reported in the monthly performance pack.

### 4.5 Actions for the coming year

The main actions for distribution network in the coming year are to:

- agree the revised targets for Kestrel Mere with the regulator before January;
- train 5 additional operators in the procedures introduced at Hartsfield;
- report progress on the actions to the operating board every quarter;
- close the 210 outstanding work orders recorded for the North Vale zone;

The risk assessment for Wexcombe was reviewed in March and the control measures were found to be adequate. Results for Wexcombe have been restated for March after a meter calibration error was found. The local authority was informed of the work at Wexcombe in advance, and no objections were received. The budget for this activity at Pellow Down was underspent by 89 per cent because of delays in procurement.

## 5 Leakage and Non-Revenue Water

1,431 meters were replaced at the end of their service life under the planned programme. The values reported for Thornby are provisional for January and will be confirmed in the first quarterly return. The long-term trend at Fenwick Moor shows a gradual improvement of about 7.6 per cent a year. Night use allowances were updated in February using the results of the household consumption monitor. Leakage in the Central zone fell to 6.5 megalitres per day, the lowest value since metering began.

A review of the data for Netherby found no evidence of drift in the 79 instruments used for the returns. Wexcombe and Longridge together account for 85 per cent of the total for the network. The long-term trend at Kestrel Mere shows a gradual improvement of about 2.9 per cent a year. The standby generator at Quarry Lane was tested monthly and ran for 0.4 hours during the year. Seasonal variation at Hartsfield was more pronounced than usual because of the dry spell in late April.

### 5.1 Method

The water balance closes within 3.5 per cent, which is inside the tolerance set by the regulator. The asset register entry for Pellow Down was updated in December to reflect the equipment now installed. Night use allowances were updated in May using the results of the household consumption monitor. Samples from Marlpit were analysed by the accredited laboratory within 8 hours of collection. Leakage is calculated from the minimum night flows in district metered areas, adjusted for legitimate night use. No adjustment has been made for the transfer of the Upland zone boundary, which affects fewer than 89 properties.

Seasonal variation at Kestrel Mere was more pronounced than usual because of the dry spell in late August. Leakage is calculated from the minimum night flows in district metered areas,

adjusted for legitimate night use. The local authority was informed of the work at Thornby in advance, and no objections were received. District metered area meters at Coldharbour were verified in situ with an insertion probe. Samples from Oakhanger were analysed by the accredited laboratory within 17 hours of collection. The water balance closes within 3.9 per cent, which is inside the tolerance set by the regulator.

The risk assessment for Ashby Weir was reviewed in October and the control measures were found to be adequate. Leakage is calculated from the minimum night flows in district metered areas, adjusted for legitimate night use. Night use allowances were updated in March using the results of the household consumption monitor. The local authority was informed of the work at Longridge in advance, and no objections were received. The values reported for Thornby are provisional for December and will be confirmed in the first quarterly return.

The standby generator at Coldharbour was tested monthly and ran for 6.2 hours during the year. Training on the new procedure was given to 8 operators from Ashby Weir and Quarry Lane. A review of the data for Longridge found no evidence of drift in the 199 instruments used for the returns. Leakage is calculated from the minimum night flows in district metered areas, adjusted for legitimate night use. Night use allowances were updated in March using the results of the household consumption monitor. The water balance closes within 1.0 per cent, which is inside the tolerance set by the regulator.

Night use allowances were updated in June using the results of the household consumption monitor. The values reported for Coldharbour are provisional for April and will be confirmed in the first quarterly return. District metered area meters at Pellow Down were verified in situ with an insertion probe. The water balance closes within 6.0 per cent, which is inside the tolerance set by the regulator. The long-term trend at Upwell shows a gradual improvement of about 7.7 per cent a year. The site visit to Kestrel Mere in June confirmed that the readings match the local display.

### 5.2 Results by zone

The Upland zone remains the main contributor to total leakage because of its older mains. A review of the data for Ashby Weir found no evidence of drift in the 180 instruments used for the returns. An independent check of the Central zone returns in December found 12 minor transcription errors, now corrected. No adjustment has been made for the transfer of the Central zone boundary, which affects fewer than 188 properties. Leakage in the Upland zone fell to 6.9 megalitres per day, the lowest value since metering began.

Leakage in the Coastal zone fell to 0.7 megalitres per day, the lowest value since metering began. The asset register entry for Yarrowby was updated in January to reflect the equipment now installed. The supplier of the equipment at Marlpit confirmed that support will continue for at least 2 years. The Westmoor zone remains the main contributor to total leakage because of its older mains. Leakage per property in the Central zone is now below the network average for the first time. Gallow Hill and Marlpit together account for 30 per cent of the total for the network.

Remote access to the controls at Yarrowby was enabled in June after the security review. The rise in leakage at Upwell in May was traced to a failed joint on a trunk main. Leakage per property in the Central zone is now below the network average for the first time. Leakage in the Coastal zone fell to 5.9 megalitres per day, the lowest value since metering began. Netherby and Oakhanger together account for 55 per cent of the total for the network. At Ashby Weir the change compared with the previous year, 0.9 per cent, is within the uncertainty of the measurement.

Spare parts for the equipment at Longridge are now held at the North Vale depot to shorten repair times. The Eastgate zone remains the main contributor to total leakage because of its older

mains. Two inspections at Oakhanger found no defects, and the next inspection is scheduled for March. An independent check of the Eastgate zone returns in October found 3 minor transcription errors, now corrected. Leakage per property in the Upland zone is now below the network average for the first time.

| Month | Yarrowby | Netherby | Marlpit | Thornby |
|---|---|---|---|---|
| January | 21.4 | 53.3 | 47.9 | 30.6 |
| February | 48.0 | 29.0 | 48.6 | 12.3 |
| March | 4.4 | 14.7 | 9.4 | 52.7 |
| April | 59.1 | 35.8 | 46.0 | 53.5 |
| May | 40.1 | 4.8 | 53.7 | 48.2 |
| June | 35.9 | 37.0 | 2.2 | 35.2 |
| July | 15.9 | 24.9 | 2.1 | 41.8 |
| August | 15.0 | 29.2 | 14.1 | 14.0 |
| September | 2.8 | 14.7 | 5.4 | 33.3 |
| October | 11.8 | 14.0 | 48.3 | 49.0 |
| November | 39.5 | 15.4 | 1.8 | 21.0 |
| December | 49.5 | 32.2 | 34.8 | 32.7 |

Table 5.1 – Monthly values for leakage and non-revenue water, selected sites

Values are monthly averages from the operational database; provisional values are included for December.

### 5.3 Active leakage control

Correlator surveys in March located several small leaks on communication pipes in the Westmoor zone. Acoustic loggers were deployed permanently in 70 district metered areas around Ravensgill. The control room logged 217 alarms from Longridge in March, most of them cleared without a site visit. Detection teams surveyed 4,116 kilometres of main and found 36 leaks that had not been reported. The budget for this activity at Ashby Weir was underspent by 43 per cent because of delays in procurement. 125 work orders were raised for Hartsfield during the year, compared with 24,275 across the network.

Average time from detection to repair of non-visible leaks was 13 days. The local authority was informed of the work at Ravensgill in advance, and no objections were received. Acoustic loggers were deployed permanently in 59 district metered areas around Wexcombe. Following a near miss at Yarrowby in February, the access arrangements were changed. The control room logged 130 alarms from Fenwick Moor in October, most of them cleared without a site visit.

Improvements at Hartsfield are expected to save about 36,675 pounds a year in operating costs. Detection teams surveyed 14,061 kilometres of main and found 111 leaks that had not been reported. Acoustic loggers were deployed permanently in 221 district metered areas around Thornby. Data from Kestrel Mere for the first 13 days of August were rejected by the validation rules and infilled. Staff from the Upland depot supported the work at Coldharbour during the busy period in July. Correlator surveys in December located several small leaks on communication pipes in the Westmoor zone.

Remote access to the controls at Saltcote was enabled in April after the security review. Acoustic loggers were deployed permanently in 28 district metered areas around Wexcombe. The values reported for Upwell are provisional for July and will be confirmed in the first quarterly return. Spare parts for the equipment at Pellow Down are now held at the North Vale depot to shorten repair times. Correlator surveys in January located several small leaks on communication pipes in the Eastgate zone.

Gallow Hill and Upwell together account for 85 per cent of the total for the network. Following a near miss at Thornby in September, the access arrangements were changed. Acoustic loggers were deployed permanently in 176 district metered areas around Upwell. Correlator surveys in June located several small leaks on communication pipes in the Riverside zone. No adjustment

has been made for the transfer of the Upland zone boundary, which affects fewer than 66 properties. Detection teams surveyed 18,900 kilometres of main and found 219 leaks that had not been reported.

The values reported for Longridge are provisional for July and will be confirmed in the first quarterly return. Spare parts for the equipment at Kestrel Mere are now held at the Westmoor depot to shorten repair times. The value for Longridge is the highest in the Upland zone, reflecting the size of the area it serves.[2]

### 5.4 Customer metering

Smart meter trials in the Upland zone identified continuous flows at 35 properties. Staff from the Coastal depot supported the work at Coldharbour during the busy period in May. Spare parts for the equipment at Oakhanger are now held at the Eastgate depot to shorten repair times. Meter penetration among household customers reached 34 per cent at the end of the year. The control room logged 62 alarms from Fenwick Moor in September, most of them cleared without a site visit. 42,023 meters were replaced at the end of their service life under the planned programme.

Smart meter trials in the Riverside zone identified continuous flows at 207 properties. Meter penetration among household customers reached 49 per cent at the end of the year. Data from Kestrel Mere for the first 15 days of June were rejected by the validation rules and infilled. Results for Wexcombe have been restated for June after a meter calibration error was found. The asset register entry for Hartsfield was updated in September to reflect the equipment now installed.

Operational staff at Quarry Lane reported that the procedures revised in November were easy to apply on night shifts. The control room logged 155 alarms from Upwell in February, most of them cleared without a site visit. Remote access to the controls at Pellow Down was enabled in March after the security review. Smart meter trials in the North Vale zone identified continuous flows at 39 properties. 10,681 meters were replaced at the end of their service life under the planned programme.

Operational staff at Oakhanger reported that the procedures revised in September were easy to apply on night shifts. Meter penetration among household customers reached 29 per cent at the end of the year. The trend at Kestrel Mere reversed in March after the works described in the previous section. At Coldharbour the change compared with the previous year, 8.3 per cent, is within the uncertainty of the measurement. Smart meter trials in the Coastal zone identified continuous flows at 139 properties.

Data from Coldharbour for the first 7 days of July were rejected by the validation rules and infilled. 118 work orders were raised for Upwell during the year, compared with 40,631 across the network. Smart meter trials in the North Vale zone identified continuous flows at 39 properties. The long-term trend at Netherby shows a gradual improvement of about 7.7 per cent a year. Meter penetration among household customers reached 24 per cent at the end of the year.

| Month | Ashby Weir | Wexcombe | Oakhanger | Thornby | Pellow Down |
|---|---|---|---|---|---|
| January | 39.3 | 38.5 | 24.2 | 69.2 | 31.4 |
| February | 77.6 | 51.9 | 76.0 | 57.9 | 75.5 |
| March | 92.2 | 75.2 | 65.0 | 22.1 | 88.8 |
| April | 85.8 | 19.6 | 17.9 | 96.8 | 70.5 |
| May | 66.3 | 91.2 | 28.7 | 85.5 | 23.9 |
| June | 13.9 | 85.0 | 13.6 | 25.5 | 56.6 |
| July | 87.7 | 21.3 | 72.4 | 17.3 | 78.2 |
| August | 12.0 | 67.4 | 50.6 | 84.7 | 34.4 |

[2] The asset register entry for Longridge was updated in December to reflect the equipment now installed. Staff from the Westmoor depot supported the work at Ashby Weir during the busy period in September. Following a near miss at Hartsfield in March, the access arrangements were changed.

| Month | Ashby Weir | Wexcombe | Oakhanger | Thornby | Pellow Down |
|---|---|---|---|---|---|
| September | 95.2 | 75.3 | 57.2 | 69.6 | 70.6 |
| October | 84.2 | 12.6 | 81.2 | 61.1 | 57.0 |
| November | 36.6 | 48.8 | 31.6 | 10.3 | 10.4 |
| December | 16.8 | 38.2 | 50.0 | 63.8 | 79.1 |

Table 5.2 – Monthly indicators for customer metering, further sites

Indicator values as reported in the monthly performance pack.

### 5.5 Actions for the coming year

The main actions for leakage and non-revenue water in the coming year are to:

- publish a summary of the results for the Westmoor zone on the website;
- review the spare parts held at the Riverside depot for Gallow Hill and Ravensgill;
- extend the monitoring programme to 19 further sites in the Upland zone;
- report progress on the actions to the operating board every quarter;

The budget for this activity at Quarry Lane was underspent by 26 per cent because of delays in procurement. Two inspections at Quarry Lane found no defects, and the next inspection is scheduled for November. Pellow Down and Quarry Lane together account for 25 per cent of the total for the network. The long-term trend at Gallow Hill shows a gradual improvement of about 7.3 per cent a year. Data from Saltcote for the first 16 days of November were rejected by the validation rules and infilled.

## 6 Energy Use and Carbon

The energy intensity of treated water supplied was 2.8 kilowatt hours per cubic metre. Operational emissions were estimated at 33,385 tonnes of carbon dioxide equivalent. The site visit to Coldharbour in April confirmed that the readings match the local display. The hydro turbine at Oakhanger generated 24,711 megawatt hours, slightly less than in the previous year.

A review of the data for Netherby found no evidence of drift in the 25 instruments used for the returns. The long-term trend at Saltcote shows a gradual improvement of about 1.7 per cent a year. The control room logged 214 alarms from Wexcombe in December, most of them cleared without a site visit. At Hartsfield the change compared with the previous year, 9.5 per cent, is within the uncertainty of the measurement. The asset register entry for Thornby was updated in June to reflect the equipment now installed. Results for Gallow Hill have been restated for March after a meter calibration error was found.

### 6.1 Pumping energy

Energy use at Quarry Lane rose in September because the gravity transfer was out of service. Operational staff at Yarrowby reported that the procedures revised in July were easy to apply on night shifts. Remote access to the controls at Yarrowby was enabled in November after the security review. The control room logged 87 alarms from Netherby in February, most of them cleared without a site visit. The pump refurbishment at Ravensgill restored the efficiency of the duty pump to near its design value.

The asset register entry for Hartsfield was updated in May to reflect the equipment now installed. Performance at Hartsfield was affected for 8 days by a power failure on the distribution network in October. Total electricity consumption was 23,946 megawatt hours, most of it for raw water and booster pumping. Variable speed drives were fitted to the booster pumps serving the Upland zone. The supplier of the equipment at Coldharbour confirmed that support will continue for at least 10 years.

Seasonal variation at Netherby was more pronounced than usual because of the dry spell in late April. The pump refurbishment at Fenwick Moor restored the efficiency of the duty pump to near its design value. Coldharbour and Saltcote together account for 34 per cent of the total for the network. Variable speed drives were fitted to the booster pumps serving the Riverside zone. Total electricity consumption was 33,713 megawatt hours, most of it for raw water and booster pumping. The figures for Coldharbour include an estimate for October, when the meter at the outlet was out of service for 11 days.

Total electricity consumption was 27,152 megawatt hours, most of it for raw water and booster pumping. Results for Marlpit have been restated for December after a meter calibration error was found. Variable speed drives were fitted to the booster pumps serving the Central zone. The value for Wexcombe is the highest in the Coastal zone, reflecting the size of the area it serves. At Saltcote the change compared with the previous year, 3.9 per cent, is within the uncertainty of the measurement. Energy use at Hartsfield rose in August because the gravity transfer was out of service.

Oakhanger and Ashby Weir together account for 55 per cent of the total for the network. Variable speed drives were fitted to the booster pumps serving the Riverside zone. Staff from the Westmoor depot supported the work at Pellow Down during the busy period in August. Energy use at Longridge rose in March because the gravity transfer was out of service. The value for Quarry Lane is the highest in the North Vale zone, reflecting the size of the area it serves.

### 6.2 Renewable generation

The hydro turbine at Wexcombe generated 27,731 megawatt hours, slightly less than in the previous year. Remote access to the controls at Pellow Down was enabled in November after the security review. The long-term trend at Wexcombe shows a gradual improvement of about 6.1 per cent a year. The turbine at Netherby was out of service for 9 days in December for bearing replacement. The risk assessment for Pellow Down was reviewed in June and the control measures were found to be adequate. Solar panels on the roof of the Oakhanger works produced 45 per cent of the site consumption.

The turbine at Oakhanger was out of service for 7 days in October for bearing replacement. Solar panels on the roof of the Netherby works produced 62 per cent of the site consumption. Improvements at Saltcote are expected to save about 19,843 pounds a year in operating costs. Remote access to the controls at Thornby was enabled in June after the security review. The hydro turbine at Pellow Down generated 12,474 megawatt hours, slightly less than in the previous year. Spare parts for the equipment at Wexcombe are now held at the Riverside depot to shorten repair times.

The turbine at Coldharbour was out of service for 16 days in July for bearing replacement. Improvements at Upwell are expected to save about 6,332 pounds a year in operating costs. Solar panels on the roof of the Fenwick Moor works produced 42 per cent of the site consumption. The hydro turbine at Yarrowby generated 10,702 megawatt hours, slightly less than in the previous year. The site visit to Yarrowby in January confirmed that the readings match the local display. Results for Kestrel Mere have been restated for February after a meter calibration error was found.

Solar panels on the roof of the Upwell works produced 19 per cent of the site consumption. Two inspections at Oakhanger found no defects, and the next inspection is scheduled for December. A contractor carried out the work at Upwell under the framework agreement, finishing 9 days ahead of plan. The hydro turbine at Ashby Weir generated 33,596 megawatt hours, slightly less than in the previous year. Seasonal variation at Longridge was more pronounced than usual

because of the dry spell in late June. The turbine at Ashby Weir was out of service for 9 days in July for bearing replacement.

| Month | Saltcote | Wexcombe | Hartsfield | Quarry Lane |
|---|---|---|---|---|
| January | 21.3 | 5.3 | 48.4 | 13.8 |
| February | 24.7 | 48.0 | 54.6 | 7.5 |
| March | 18.2 | 37.6 | 30.6 | 5.1 |
| April | 23.1 | 2.1 | 24.1 | 47.1 |
| May | 35.1 | 32.7 | 39.1 | 21.2 |
| June | 13.1 | 24.6 | 24.3 | 23.4 |
| July | 15.7 | 52.1 | 5.6 | 53.7 |
| August | 4.6 | 50.7 | 50.8 | 43.7 |
| September | 34.5 | 49.8 | 6.8 | 37.7 |
| October | 48.5 | 49.2 | 40.1 | 32.1 |
| November | 47.4 | 56.3 | 35.4 | 14.4 |
| December | 56.8 | 43.2 | 11.2 | 7.1 |

Table 6.1 – Monthly values for energy use and carbon, selected sites

Values are monthly averages from the operational database; provisional values are included for December.

### 6.3 Energy intensity

The budget for this activity at Fenwick Moor was underspent by 91 per cent because of delays in procurement. The intensity at Thornby fell after the transfer main to Yarrowby was brought back into service. Intensity is highest at the borehole sources, where water is pumped from depth before treatment. The energy intensity of treated water supplied was 0.7 kilowatt hours per cubic metre. Operational staff at Pellow Down reported that the procedures revised in May were easy to apply on night shifts. Results for Coldharbour were in line with the previous year, with no change in the operating regime after May.

Spare parts for the equipment at Oakhanger are now held at the Westmoor depot to shorten repair times. The energy intensity of treated water supplied was 4.4 kilowatt hours per cubic metre. A contractor carried out the work at Fenwick Moor under the framework agreement, finishing 16 days ahead of plan. Intensity is highest at the borehole sources, where water is pumped from depth before treatment. Operational staff at Saltcote reported that the procedures revised in December were easy to apply on night shifts. The intensity at Saltcote fell after the transfer main to Wexcombe was brought back into service.

The risk assessment for Yarrowby was reviewed in July and the control measures were found to be adequate. The local authority was informed of the work at Fenwick Moor in advance, and no objections were received. The energy intensity of treated water supplied was 6.4 kilowatt hours per cubic metre. The intensity at Ashby Weir fell after the transfer main to Wexcombe was brought back into service. Spare parts for the equipment at Marlpit are now held at the Upland depot to shorten repair times.

Data from Ashby Weir for the first 17 days of November were rejected by the validation rules and infilled. The control room logged 30 alarms from Saltcote in October, most of them cleared without a site visit. The energy intensity of treated water supplied was 2.1 kilowatt hours per cubic metre. Intensity is highest at the borehole sources, where water is pumped from depth before treatment. Results for Coldharbour have been restated for July after a meter calibration error was found.

Staff from the Coastal depot supported the work at Upwell during the busy period in March. Intensity is highest at the borehole sources, where water is pumped from depth before treatment. The method used for Wexcombe differs from that used for Kestrel Mere, so the two values are not directly comparable. A study of options for Oakhanger is planned, with a report due to the

operating board in February. The intensity at Thornby fell after the transfer main to Quarry Lane was brought back into service.

### 6.4 Carbon accounting

Fleet emissions fell by 87 per cent after electric vans were introduced in the Eastgate depot. A study of options for Gallow Hill is planned, with a report due to the operating board in September. Detail for Wexcombe and the other sites in the Eastgate zone is held in the operational database. Emission factors for purchased electricity follow the national reporting guidance for the year. Remote access to the controls at Yarrowby was enabled in November after the security review.

Emission factors for purchased electricity follow the national reporting guidance for the year. Following a near miss at Gallow Hill in September, the access arrangements were changed. Samples from Ashby Weir were analysed by the accredited laboratory within 10 hours of collection. A contractor carried out the work at Fenwick Moor under the framework agreement, finishing 14 days ahead of plan. Operational emissions were estimated at 18,942 tonnes of carbon dioxide equivalent. Process emissions from sludge treatment at Marlpit were estimated from the volume of sludge produced.

Operational emissions were estimated at 28,286 tonnes of carbon dioxide equivalent. Following a near miss at Ashby Weir in March, the access arrangements were changed. The trend at Kestrel Mere reversed in October after the works described in the previous section. Fleet emissions fell by 73 per cent after electric vans were introduced in the Westmoor depot. The values reported for Ashby Weir are provisional for March and will be confirmed in the first quarterly return. Process emissions from sludge treatment at Netherby were estimated from the volume of sludge produced.

Improvements at Hartsfield are expected to save about 38,721 pounds a year in operating costs. Process emissions from sludge treatment at Wexcombe were estimated from the volume of sludge produced. Data from Netherby for the first 19 days of September were rejected by the validation rules and infilled. Fleet emissions fell by 33 per cent after electric vans were introduced in the Coastal depot. The control room logged 171 alarms from Ashby Weir in October, most of them cleared without a site visit.

The local authority was informed of the work at Kestrel Mere in advance, and no objections were received. Operational emissions were estimated at 34,644 tonnes of carbon dioxide equivalent. Process emissions from sludge treatment at Yarrowby were estimated from the volume of sludge produced. Fleet emissions fell by 54 per cent after electric vans were introduced in the Westmoor depot. An independent check of the Eastgate zone returns in September found 19 minor transcription errors, now corrected. The method used for Fenwick Moor differs from that used for Gallow Hill, so the two values are not directly comparable.

| Month | Upwell | Wexcombe | Ashby Weir | Longridge | Gallow Hill |
|---|---|---|---|---|---|
| January | 88.3 | 82.3 | 10.4 | 51.8 | 98.5 |
| February | 77.9 | 76.5 | 34.3 | 15.4 | 49.6 |
| March | 25.7 | 93.9 | 69.7 | 64.6 | 50.6 |
| April | 92.5 | 56.4 | 68.1 | 16.9 | 97.7 |
| May | 39.5 | 74.3 | 78.1 | 53.0 | 54.1 |
| June | 68.9 | 81.0 | 85.7 | 87.0 | 71.1 |
| July | 71.5 | 40.7 | 10.1 | 82.1 | 97.9 |
| August | 59.5 | 86.2 | 83.2 | 11.1 | 43.7 |
| September | 13.5 | 19.5 | 31.1 | 11.7 | 45.6 |
| October | 43.0 | 42.8 | 97.6 | 45.6 | 79.3 |
| November | 23.8 | 73.1 | 60.3 | 45.6 | 93.2 |
| December | 91.9 | 65.0 | 95.2 | 95.4 | 69.2 |

Table 6.2 – Monthly indicators for carbon accounting, further sites

Indicator values as reported in the monthly performance pack.

### 6.5 Actions for the coming year

The main actions for energy use and carbon in the coming year are to:

- review the spare parts held at the Westmoor depot for Ravensgill and Thornby;
- train 11 additional operators in the procedures introduced at Coldharbour;
- report progress on the actions to the operating board every quarter;
- publish a summary of the results for the Eastgate zone on the website;

The standby generator at Pellow Down was tested monthly and ran for 9.4 hours during the year. The risk assessment for Saltcote was reviewed in December and the control measures were found to be adequate. Two inspections at Pellow Down found no defects, and the next inspection is scheduled for January. Samples from Thornby were analysed by the accredited laboratory within 13 hours of collection. Samples from Wexcombe were analysed by the accredited laboratory within 7 hours of collection.

## 7 Asset Condition and Maintenance

The value for Netherby is the highest in the Central zone, reflecting the size of the area it serves. Condition grades are recorded on a five-point scale and stored in the asset register. 160 work orders were raised for Wexcombe during the year, compared with 14,422 across the network. Most outstanding work orders relate to minor civil repairs at small booster stations. The planned maintenance backlog was reduced to 122 work orders by the end of the year.

Demand from the North Vale zone was 9.7 per cent higher in April than in the same month a year earlier. The supplier of the equipment at Oakhanger confirmed that support will continue for at least 3 years. The site visit to Coldharbour in September confirmed that the readings match the local display. Operational staff at Ravensgill reported that the procedures revised in October were easy to apply on night shifts. No adjustment has been made for the transfer of the Upland zone boundary, which affects fewer than 150 properties.

### 7.1 Inspection programme

The control room logged 198 alarms from Oakhanger in July, most of them cleared without a site visit. The values reported for Pellow Down are provisional for December and will be confirmed in the first quarterly return. 21,945 asset inspections were completed, 28 per cent of the planned number for the year. The planning assumption for Oakhanger is unchanged at 3.8 per cent above the long-term average. Condition grades are recorded on a five-point scale and stored in the asset register.

Improvements at Kestrel Mere are expected to save about 47,690 pounds a year in operating costs. The budget for this activity at Wexcombe was underspent by 58 per cent because of delays in procurement. Inspections at Marlpit were rescheduled to August to avoid the period of peak demand. 2,607 asset inspections were completed, 78 per cent of the planned number for the year. Condition grades are recorded on a five-point scale and stored in the asset register. The trend at Hartsfield reversed in September after the works described in the previous section.

Condition grades are recorded on a five-point scale and stored in the asset register. 30,809 asset inspections were completed, 65 per cent of the planned number for the year. A contractor carried out the work at Hartsfield under the framework agreement, finishing 16 days ahead of plan. Samples from Hartsfield were analysed by the accredited laboratory within 19 hours of collection. Seasonal variation at Gallow Hill was more pronounced than usual because of the

dry spell in late June. Inspections at Pellow Down were rescheduled to April to avoid the period of peak demand.

The value for Marlpit is the highest in the Coastal zone, reflecting the size of the area it serves. A contractor carried out the work at Gallow Hill under the framework agreement, finishing 3 days ahead of plan. Inspections at Gallow Hill were rescheduled to February to avoid the period of peak demand. Condition grades are recorded on a five-point scale and stored in the asset register. The long-term trend at Kestrel Mere shows a gradual improvement of about 7.7 per cent a year.

### 7.2 Service reservoirs

170 work orders were raised for Pellow Down during the year, compared with 43,597 across the network. The internal inspection of the service reservoir at Thornby found minor spalling of the roof slab. Roof membranes were renewed at 2 service reservoirs, including the two cells at Quarry Lane. Operational staff at Marlpit reported that the procedures revised in January were easy to apply on night shifts. Access hatches at Wexcombe were replaced with lockable covers fitted with intruder alarms. A review of the data for Netherby found no evidence of drift in the 69 instruments used for the returns.

The internal inspection of the service reservoir at Hartsfield found minor spalling of the roof slab. The values reported for Hartsfield are provisional for June and will be confirmed in the first quarterly return. Access hatches at Wexcombe were replaced with lockable covers fitted with intruder alarms. The trend at Pellow Down reversed in February after the works described in the previous section. A contractor carried out the work at Netherby under the framework agreement, finishing 4 days ahead of plan.

Night work at Saltcote was limited to 6 shifts to reduce noise for neighbouring residents. Roof membranes were renewed at 14 service reservoirs, including the two cells at Ashby Weir. The internal inspection of the service reservoir at Saltcote found minor spalling of the roof slab. 28 work orders were raised for Yarrowby during the year, compared with 30,099 across the network. The site visit to Oakhanger in February confirmed that the readings match the local display.

Access hatches at Netherby were replaced with lockable covers fitted with intruder alarms. Operational staff at Quarry Lane reported that the procedures revised in April were easy to apply on night shifts. Roof membranes were renewed at 17 service reservoirs, including the two cells at Ashby Weir. Demand from the Central zone was 3.3 per cent higher in December than in the same month a year earlier. A contractor carried out the work at Yarrowby under the framework agreement, finishing 14 days ahead of plan.

| Month | Upwell | Pellow Down | Ravensgill | Marlpit |
|---|---|---|---|---|
| January | 47.6 | 41.4 | 47.3 | 46.2 |
| February | 5.1 | 26.6 | 25.6 | 50.1 |
| March | 18.7 | 17.0 | 14.0 | 4.8 |
| April | 40.6 | 37.9 | 38.8 | 57.6 |
| May | 25.8 | 8.0 | 5.0 | 3.0 |
| June | 19.6 | 13.0 | 22.0 | 12.8 |
| July | 33.1 | 1.0 | 25.8 | 3.4 |
| August | 26.4 | 59.6 | 35.9 | 38.8 |
| September | 50.5 | 16.2 | 39.6 | 14.9 |
| October | 6.7 | 26.6 | 22.4 | 19.5 |
| November | 43.5 | 12.8 | 58.9 | 19.4 |
| December | 58.4 | 49.9 | 6.6 | 15.6 |

Table 7.1 – Monthly values for asset condition and maintenance, selected sites

Values are monthly averages from the operational database; provisional values are included for December.

### 7.3 Pumps and motors

The duty pump at Marlpit was rewound in February after an insulation fault was detected. Mean time between failures for booster pumps improved to 45,934 hours. The asset register entry for Yarrowby was updated in February to reflect the equipment now installed. The method used for Upwell differs from that used for Pellow Down, so the two values are not directly comparable. Vibration monitoring was extended to 187 pumps, and alarms are now reported to the control room. Performance at Netherby was affected for 8 days by a power failure on the distribution network in March.

The duty pump at Wexcombe was rewound in May after an insulation fault was detected. Seasonal variation at Yarrowby was more pronounced than usual because of the dry spell in late April. Mean time between failures for booster pumps improved to 28,560 hours. Samples from Fenwick Moor were analysed by the accredited laboratory within 9 hours of collection. The control room logged 99 alarms from Saltcote in August, most of them cleared without a site visit.

The site visit to Ashby Weir in April confirmed that the readings match the local display. The local authority was informed of the work at Marlpit in advance, and no objections were received. 95 work orders were raised for Longridge during the year, compared with 15,501 across the network. Mean time between failures for booster pumps improved to 13,398 hours. The duty pump at Hartsfield was rewound in October after an insulation fault was detected.

The local authority was informed of the work at Netherby in advance, and no objections were received. Mean time between failures for booster pumps improved to 29,507 hours. The duty pump at Quarry Lane was rewound in August after an insulation fault was detected. Vibration monitoring was extended to 42 pumps, and alarms are now reported to the control room. Following a near miss at Longridge in July, the access arrangements were changed. The trend at Ravensgill reversed in April after the works described in the previous section.

Vibration monitoring was extended to 141 pumps, and alarms are now reported to the control room. Mean time between failures for booster pumps improved to 29,281 hours. An independent check of the Riverside zone returns in October found 19 minor transcription errors, now corrected. The duty pump at Oakhanger was rewound in January after an insulation fault was detected. Night work at Ravensgill was limited to 14 shifts to reduce noise for neighbouring residents. Performance at Ashby Weir was affected for 6 days by a power failure on the distribution network in July.

Samples from Ashby Weir were analysed by the accredited laboratory within 19 hours of collection. The risk assessment for Gallow Hill was reviewed in January and the control measures were found to be adequate. The local authority was informed of the work at Oakhanger in advance, and no objections were received.[3]

### 7.4 Maintenance backlog

Spare parts for the equipment at Saltcote are now held at the Westmoor depot to shorten repair times. The value for Kestrel Mere is the highest in the Upland zone, reflecting the size of the area it serves. Staff from the Riverside depot supported the work at Yarrowby during the busy period in January. Reactive work fell to 24 per cent of total maintenance hours, compared with the previous year. Most outstanding work orders relate to minor civil repairs at small booster stations.

[3] The control room logged 47 alarms from Ravensgill in March, most of them cleared without a site visit. The method used for Quarry Lane differs from that used for Netherby, so the two values are not directly comparable. The site visit to Wexcombe in January confirmed that the readings match the local display.

The planned maintenance backlog was reduced to 30 work orders by the end of the year. The asset register entry for Saltcote was updated in November to reflect the equipment now installed. Data from Pellow Down for the first 4 days of November were rejected by the validation rules and infilled. Improvements at Hartsfield are expected to save about 24,329 pounds a year in operating costs. Most outstanding work orders relate to minor civil repairs at small booster stations.

An independent check of the Riverside zone returns in March found 11 minor transcription errors, now corrected. Samples from Upwell were analysed by the accredited laboratory within 15 hours of collection. Reactive work fell to 44 per cent of total maintenance hours, compared with the previous year. The standby generator at Oakhanger was tested monthly and ran for 7.8 hours during the year. The planned maintenance backlog was reduced to 94 work orders by the end of the year.

The planned maintenance backlog was reduced to 204 work orders by the end of the year. Most outstanding work orders relate to minor civil repairs at small booster stations. Reactive work fell to 23 per cent of total maintenance hours, compared with the previous year. Demand from the Upland zone was 8.8 per cent higher in December than in the same month a year earlier. The risk assessment for Pellow Down was reviewed in November and the control measures were found to be adequate. Results for Netherby were in line with the previous year, with no change in the operating regime after February.

Most outstanding work orders relate to minor civil repairs at small booster stations. Following a near miss at Yarrowby in June, the access arrangements were changed. Data from Kestrel Mere for the first 8 days of October were rejected by the validation rules and infilled. Reactive work fell to 21 per cent of total maintenance hours, compared with the previous year. 56 work orders were raised for Longridge during the year, compared with 39,764 across the network.

| Month | Thornby | Ashby Weir | Upwell | Coldharbour | Saltcote |
|---|---|---|---|---|---|
| January | 23.8 | 87.8 | 38.2 | 84.7 | 23.9 |
| February | 93.4 | 69.2 | 47.7 | 29.5 | 58.4 |
| March | 24.0 | 62.7 | 70.0 | 94.3 | 35.2 |
| April | 74.5 | 96.8 | 37.5 | 68.2 | 98.8 |
| May | 17.2 | 48.4 | 90.3 | 95.1 | 37.5 |
| June | 70.2 | 57.4 | 56.8 | 90.9 | 84.6 |
| July | 51.6 | 50.3 | 97.8 | 43.1 | 96.5 |
| August | 84.4 | 90.9 | 35.4 | 53.5 | 47.9 |
| September | 56.9 | 57.2 | 71.4 | 33.2 | 13.2 |
| October | 27.1 | 73.4 | 67.4 | 16.8 | 59.5 |
| November | 71.3 | 23.8 | 12.8 | 88.7 | 31.4 |
| December | 71.1 | 31.9 | 11.0 | 54.6 | 72.5 |

Table 7.2 – Monthly indicators for maintenance backlog, further sites

Indicator values as reported in the monthly performance pack.

### 7.5 Actions for the coming year

The main actions for asset condition and maintenance in the coming year are to:

- review the spare parts held at the Westmoor depot for Netherby and Gallow Hill;
- train 5 additional operators in the procedures introduced at Fenwick Moor;
- replace the obsolete controller at Marlpit under the asset programme;
- complete the review of operating procedures at Quarry Lane by the end of December;

Detail for Ravensgill and the other sites in the Eastgate zone is held in the operational database. A study of options for Marlpit is planned, with a report due to the operating board in March. The values reported for Gallow Hill are provisional for November and will be confirmed in the first quarterly return. Results for Pellow Down were in line with the previous year, with no change in

the operating regime after June. Results for Oakhanger have been restated for September after a meter calibration error was found.

## 8 Customer Service

Written complaints totalled 124, and 78 per cent were resolved at the first response. The value for Coldharbour is the highest in the Westmoor zone, reflecting the size of the area it serves. The control room logged 46 alarms from Longridge in March, most of them cleared without a site visit. 90 properties experienced an unplanned interruption of more than three hours. The most common cause of complaint was the time taken to reinstate footpaths after repairs.

Results for Yarrowby were in line with the previous year, with no change in the operating regime after January. Ravensgill and Coldharbour together account for 19 per cent of the total for the network. Fenwick Moor and Longridge together account for 73 per cent of the total for the network. Detail for Pellow Down and the other sites in the Coastal zone is held in the operational database. Two inspections at Fenwick Moor found no defects, and the next inspection is scheduled for December. Staff from the Coastal depot supported the work at Longridge during the busy period in February.

### 8.1 Customer contacts

Calls peaked in December after the burst at Marlpit interrupted supplies to part of the North Vale zone. The risk assessment for Longridge was reviewed in April and the control measures were found to be adequate. Detail for Kestrel Mere and the other sites in the Coastal zone is held in the operational database. Contacts about water quality fell by 13 per cent compared with the previous year. The contact centre handled 13,497 customer contacts, most of them about billing and meter readings. The standby generator at Fenwick Moor was tested monthly and ran for 4.7 hours during the year.

Calls peaked in May after the burst at Saltcote interrupted supplies to part of the Central zone. A review of the data for Yarrowby found no evidence of drift in the 103 instruments used for the returns. Contacts about water quality fell by 14 per cent compared with the previous year. The contact centre handled 3,902 customer contacts, most of them about billing and meter readings. Results for Ashby Weir were in line with the previous year, with no change in the operating regime after November. Night work at Marlpit was limited to 6 shifts to reduce noise for neighbouring residents.

Calls peaked in March after the burst at Longridge interrupted supplies to part of the Eastgate zone. The contact centre handled 27,123 customer contacts, most of them about billing and meter readings. 225 work orders were raised for Upwell during the year, compared with 34,300 across the network. The local authority was informed of the work at Thornby in advance, and no objections were received. Remote access to the controls at Wexcombe was enabled in March after the security review.

Contacts about water quality fell by 20 per cent compared with the previous year. A contractor carried out the work at Kestrel Mere under the framework agreement, finishing 8 days ahead of plan. The risk assessment for Marlpit was reviewed in August and the control measures were found to be adequate. The planning assumption for Oakhanger is unchanged at 5.9 per cent above the long-term average. The contact centre handled 7,955 customer contacts, most of them about billing and meter readings.

The contact centre handled 12,469 customer contacts, most of them about billing and meter readings. 48 work orders were raised for Netherby during the year, compared with 28,109 across

the network. The figures for Longridge include an estimate for October, when the meter at the outlet was out of service for 12 days. The site visit to Coldharbour in September confirmed that the readings match the local display. Calls peaked in July after the burst at Upwell interrupted supplies to part of the Westmoor zone.

### 8.2 Supply interruptions

The planning assumption for Oakhanger is unchanged at 8.5 per cent above the long-term average. The average supply interruption per property was 1.8 minutes, below the internal target. The asset register entry for Thornby was updated in January to reflect the equipment now installed. Training on the new procedure was given to 14 operators from Longridge and Saltcote. 124 properties experienced an unplanned interruption of more than three hours.

Bottled water was distributed at Fenwick Moor during the interruption in January. 165 properties experienced an unplanned interruption of more than three hours. Following a near miss at Coldharbour in April, the access arrangements were changed. Data from Oakhanger for the first 5 days of April were rejected by the validation rules and infilled. The trend at Upwell reversed in February after the works described in the previous section. The average supply interruption per property was 9.4 minutes, below the internal target.

Bottled water was distributed at Fenwick Moor during the interruption in September. 43 properties experienced an unplanned interruption of more than three hours. The risk assessment for Netherby was reviewed in July and the control measures were found to be adequate. The trend at Netherby reversed in October after the works described in the previous section. The average supply interruption per property was 2.1 minutes, below the internal target. Hartsfield and Coldharbour together account for 41 per cent of the total for the network.

Results for Yarrowby were in line with the previous year, with no change in the operating regime after January. 220 properties experienced an unplanned interruption of more than three hours. A review of the data for Thornby found no evidence of drift in the 42 instruments used for the returns. Bottled water was distributed at Hartsfield during the interruption in November. Remote access to the controls at Pellow Down was enabled in October after the security review.

| Month | Wexcombe | Kestrel Mere | Netherby | Oakhanger |
|---|---|---|---|---|
| January | 40.1 | 35.6 | 37.0 | 19.9 |
| February | 34.4 | 57.1 | 21.1 | 36.1 |
| March | 10.8 | 5.0 | 15.9 | 48.3 |
| April | 21.3 | 24.2 | 42.9 | 9.2 |
| May | 4.6 | 6.6 | 43.2 | 33.4 |
| June | 40.9 | 33.0 | 50.4 | 11.2 |
| July | 2.7 | 17.0 | 4.4 | 47.6 |
| August | 37.5 | 20.1 | 20.6 | 5.9 |
| September | 21.7 | 27.4 | 32.2 | 53.7 |
| October | 21.5 | 3.2 | 52.9 | 3.5 |
| November | 32.9 | 31.0 | 18.1 | 4.4 |
| December | 4.5 | 33.6 | 40.4 | 16.0 |

Table 8.1 – Monthly values for customer service, selected sites

Values are monthly averages from the operational database; provisional values are included for December.

### 8.3 Complaints

The value for Yarrowby is the highest in the Westmoor zone, reflecting the size of the area it serves. No complaint was referred to the independent consumer body during the year. The most common cause of complaint was the time taken to reinstate footpaths after repairs. Training on the new procedure was given to 14 operators from Marlpit and Ashby Weir. The supplier of the equipment at Thornby confirmed that support will continue for at least 3 years.

Night work at Saltcote was limited to 6 shifts to reduce noise for neighbouring residents. The supplier of the equipment at Wexcombe confirmed that support will continue for at least 16 years. The most common cause of complaint was the time taken to reinstate footpaths after repairs. No complaint was referred to the independent consumer body during the year. Data from Quarry Lane for the first 4 days of July were rejected by the validation rules and infilled.

The local authority was informed of the work at Yarrowby in advance, and no objections were received. No complaint was referred to the independent consumer body during the year. A contractor carried out the work at Pellow Down under the framework agreement, finishing 5 days ahead of plan. A study of options for Yarrowby is planned, with a report due to the operating board in December. Written complaints totalled 200, and 13 per cent were resolved at the first response.

37 work orders were raised for Coldharbour during the year, compared with 43,091 across the network. Performance at Quarry Lane was affected for 9 days by a power failure on the distribution network in September. Seasonal variation at Quarry Lane was more pronounced than usual because of the dry spell in late September. The most common cause of complaint was the time taken to reinstate footpaths after repairs. Written complaints totalled 158, and 11 per cent were resolved at the first response.

### 8.4 Actions for the coming year

The main actions for customer service in the coming year are to:

- carry out a condition survey of the equipment at Coldharbour in January;
- extend the monitoring programme to 9 further sites in the Riverside zone;
- complete the review of operating procedures at Ashby Weir by the end of July;
- report progress on the actions to the operating board every quarter;

A review of the data for Wexcombe found no evidence of drift in the 102 instruments used for the returns. The method used for Ravensgill differs from that used for Upwell, so the two values are not directly comparable. Following a near miss at Fenwick Moor in June, the access arrangements were changed. Two inspections at Longridge found no defects, and the next inspection is scheduled for March.

## 9 Water Quality Compliance

Drinking water quality is monitored at treatment works, service reservoirs and customer taps according to the sampling plan agreed with the drinking water regulator. This chapter summarises the results for 2025 and describes the events that required investigation.

### 9.1 Sampling programme

A total of 31,840 samples were taken and analysed for microbiological and chemical parameters. Sampling at customer taps follows a random daytime method, and the addresses are drawn each month by the laboratory information system. Samples at treatment works and service reservoirs are taken at fixed points that are checked for representativeness every year.

Results for Ravensgill have been restated for November after a meter calibration error was found. Spare parts for the equipment at Ravensgill are now held at the Upland depot to shorten repair times. A study of options for Yarrowby is planned, with a report due to the operating board in October. Seasonal variation at Yarrowby was more pronounced than usual because of the dry spell in late June. Detail for Pellow Down and the other sites in the Central zone is held in the operational database.

### 9.2 Turbidity at the treatment works

Turbidity of the final water is a key indicator of the performance of the treatment process. Table 9.1 gives the monthly mean turbidity in nephelometric turbidity units for the four surface water works. All monthly means are well below the standard of 1 NTU at the works outlet.

| Month | Ashby Weir | Kestrel Mere | Longridge | Ravensgill |
|---|---|---|---|---|
| January | 0.16 | 0.17 | 0.20 | 0.14 |
| February | 0.24 | 0.39 | 0.13 | 0.23 |
| March | 0.30 | 0.31 | 0.24 | 0.36 |
| April | 0.20 | 0.29 | 0.27 | 0.30 |
| May | 0.28 | 0.30 | 0.13 | 0.34 |
| June | 0.28 | 0.29 | 0.27 | 0.28 |
| July | 0.17 | 0.18 | 0.14 | 0.14 |
| August | 0.24 | 0.36 | 0.32 | 0.17 |
| September | 0.25 | 0.37 | 0.17 | 0.16 |
| October | 0.38 | 0.48 | 0.29 | 0.31 |
| November | 0.16 | 0.17 | 0.30 | 0.20 |
| December | 0.22 | 0.32 | 0.21 | 0.34 |

Table 9.1 – Monthly mean turbidity of final water (NTU)

Mean of continuous online measurements, validated against daily laboratory samples.

The higher value at Kestrel Mere in October followed a period of algal growth in the raw water reservoir, which was controlled by adjusting the draw-off level and the coagulant dose. Turbidity returned to its usual range within three weeks.

### 9.3 Compliance results

31 work orders were raised for Oakhanger during the year, compared with 18,670 across the network. A contractor carried out the work at Yarrowby under the framework agreement, finishing 5 days ahead of plan. Training on the new procedure was given to 7 operators from Hartsfield and Longridge. Seasonal variation at Oakhanger was more pronounced than usual because of the dry spell in late March. Performance at Saltcote was affected for 2 days by a power failure on the distribution network in September.

The risk assessment for Saltcote was reviewed in July and the control measures were found to be adequate. Data from Gallow Hill for the first 15 days of July were rejected by the validation rules and infilled. Detail for Yarrowby and the other sites in the North Vale zone is held in the operational database. A study of options for Oakhanger is planned, with a report due to the operating board in April. No adjustment has been made for the transfer of the Westmoor zone boundary, which affects fewer than 83 properties.

Night work at Fenwick Moor was limited to 7 shifts to reduce noise for neighbouring residents. Training on the new procedure was given to 15 operators from Quarry Lane and Thornby. Results for Hartsfield were in line with the previous year, with no change in the operating regime after July. The long-term trend at Quarry Lane shows a gradual improvement of about 1.0 per cent a year. Demand from the Central zone was 2.8 per cent higher in December than in the same month a year earlier. 168 work orders were raised for Ashby Weir during the year, compared with 8,103 across the network.

The trend at Oakhanger reversed in August after the works described in the previous section. At Coldharbour the change compared with the previous year, 9.5 per cent, is within the uncertainty of the measurement. Data from Quarry Lane for the first 2 days of December were rejected by the validation rules and infilled. Seasonal variation at Hartsfield was more pronounced than usual because of the dry spell in late September. Samples from Longridge were analysed by the accredited laboratory within 19 hours of collection. Demand from the Coastal zone was 5.8 per cent higher in May than in the same month a year earlier.

Performance at Fenwick Moor was affected for 18 days by a power failure on the distribution network in September. The site visit to Ravensgill in April confirmed that the readings match the local display. Training on the new procedure was given to 13 operators from Marlpit and Coldharbour. Two inspections at Thornby found no defects, and the next inspection is scheduled for July. Training on the new procedure was given to 2 operators from Ravensgill and Fenwick Moor.

Overall compliance with the chemical and microbiological standards at customer taps was 99.96 per cent. Of the 14 failures, nine were at single properties and were traced to domestic plumbing, and the remaining five were cleared on resampling.

**9.4 Events and investigations**

The control room logged 63 alarms from Pellow Down in July, most of them cleared without a site visit. At Netherby the change compared with the previous year, 6.2 per cent, is within the uncertainty of the measurement. Staff from the Westmoor depot supported the work at Thornby during the busy period in June. A contractor carried out the work at Marlpit under the framework agreement, finishing 5 days ahead of plan. At Hartsfield the change compared with the previous year, 3.6 per cent, is within the uncertainty of the measurement. The planning assumption for Pellow Down is unchanged at 3.3 per cent above the long-term average.

The budget for this activity at Fenwick Moor was underspent by 44 per cent because of delays in procurement. The asset register entry for Marlpit was updated in March to reflect the equipment now installed. 152 work orders were raised for Netherby during the year, compared with 29,753 across the network. Data from Oakhanger for the first 3 days of September were rejected by the validation rules and infilled. The budget for this activity at Pellow Down was underspent by 51 per cent because of delays in procurement. Detail for Wexcombe and the other sites in the North Vale zone is held in the operational database.

The control room logged 158 alarms from Netherby in April, most of them cleared without a site visit. Demand from the Eastgate zone was 1.7 per cent higher in October than in the same month a year earlier. Operational staff at Ravensgill reported that the procedures revised in September were easy to apply on night shifts. Results for Ravensgill were in line with the previous year, with no change in the operating regime after February. Training on the new procedure was given to 10 operators from Pellow Down and Yarrowby.

Remote access to the controls at Marlpit was enabled in February after the security review. A review of the data for Quarry Lane found no evidence of drift in the 173 instruments used for the returns. The asset register entry for Longridge was updated in October to reflect the equipment now installed. Operational staff at Hartsfield reported that the procedures revised in May were easy to apply on night shifts. The supplier of the equipment at Thornby confirmed that support will continue for at least 17 years.

The value for Ashby Weir is the highest in the Eastgate zone, reflecting the size of the area it serves. Samples from Ashby Weir were analysed by the accredited laboratory within 8 hours of collection. The budget for this activity at Saltcote was underspent by 41 per cent because of delays in procurement. At Fenwick Moor the change compared with the previous year, 9.4 per cent, is within the uncertainty of the measurement. The supplier of the equipment at Hartsfield confirmed that support will continue for at least 6 years.

# 10 Monitoring and Telemetry

The network is monitored by about 2,300 instruments connected to the central control system through outstations at sites and in the distribution network. This chapter describes the availability of the telemetry, the calibration programme and the main improvements made in 2025.

## 10.1 Telemetry availability

Telemetry availability, defined as the share of expected readings received by the control system, was 98.7 per cent across the year. Most data gaps were caused by radio links at remote outstations, which are affected by heavy rain and by vegetation growth along the radio path.

Improvements at Pellow Down are expected to save about 26,699 pounds a year in operating costs. A contractor carried out the work at Quarry Lane under the framework agreement, finishing 9 days ahead of plan. Performance at Kestrel Mere was affected for 6 days by a power failure on the distribution network in December. At Upwell the change compared with the previous year, 1.8 per cent, is within the uncertainty of the measurement. The risk assessment for Upwell was reviewed in October and the control measures were found to be adequate. A contractor carried out the work at Hartsfield under the framework agreement, finishing 2 days ahead of plan.

The site visit to Saltcote in May confirmed that the readings match the local display. Night work at Yarrowby was limited to 7 shifts to reduce noise for neighbouring residents. Performance at Longridge was affected for 17 days by a power failure on the distribution network in February. Coldharbour and Upwell together account for 49 per cent of the total for the network. Remote access to the controls at Hartsfield was enabled in September after the security review. The risk assessment for Ashby Weir was reviewed in February and the control measures were found to be adequate.

Data from Ashby Weir for the first 12 days of April were rejected by the validation rules and infilled. Spare parts for the equipment at Quarry Lane are now held at the Coastal depot to shorten repair times. Demand from the Coastal zone was 6.5 per cent higher in June than in the same month a year earlier. Results for Pellow Down were in line with the previous year, with no change in the operating regime after July. Night work at Netherby was limited to 9 shifts to reduce noise for neighbouring residents.

The values reported for Wexcombe are provisional for May and will be confirmed in the first quarterly return. The standby generator at Saltcote was tested monthly and ran for 1.8 hours during the year. Spare parts for the equipment at Saltcote are now held at the Riverside depot to shorten repair times. The asset register entry for Wexcombe was updated in January to reflect the equipment now installed. The figures for Longridge include an estimate for May, when the meter at the outlet was out of service for 3 days.

## 10.2 Calibration of online instruments

Online chlorine, turbidity and pH monitors at the treatment works are verified weekly against laboratory measurements and recalibrated when the difference exceeds the tolerance in the instrument procedure. Conductivity probes in the distribution network are sampled monthly at the reference points and replaced when the deviation cannot be corrected.

Conductivity readings are corrected to a reference temperature of 25 °C. The correction uses a linear temperature coefficient,

$$\kappa_{25} = \frac{\kappa_T}{1 + \alpha (T - 25)},$$

where the coefficient for the network water was determined from laboratory tests as $\alpha = -0.53$ per cent per kelvin relative to the uncorrected reading. The sign convention follows the instrument manufacturer and is applied by the control system before the data are stored.

The method used for Pellow Down differs from that used for Wexcombe, so the two values are not directly comparable. A contractor carried out the work at Gallow Hill under the framework agreement, finishing 5 days ahead of plan. The local authority was informed of the work at Pellow Down in advance, and no objections were received. The site visit to Coldharbour in March confirmed that the readings match the local display. Following a near miss at Pellow Down in June, the access arrangements were changed.

The figures for Yarrowby include an estimate for September, when the meter at the outlet was out of service for 14 days. Following a near miss at Wexcombe in December, the access arrangements were changed. No adjustment has been made for the transfer of the Westmoor zone boundary, which affects fewer than 61 properties. The planning assumption for Gallow Hill is unchanged at 2.1 per cent above the long-term average. Remote access to the controls at Thornby was enabled in May after the security review. A contractor carried out the work at Saltcote under the framework agreement, finishing 11 days ahead of plan.

Improvements at Gallow Hill are expected to save about 2,056 pounds a year in operating costs. An independent check of the Central zone returns in July found 5 minor transcription errors, now corrected. The method used for Hartsfield differs from that used for Fenwick Moor, so the two values are not directly comparable. Night work at Upwell was limited to 18 shifts to reduce noise for neighbouring residents. Operational staff at Marlpit reported that the procedures revised in August were easy to apply on night shifts.

Remote access to the controls at Fenwick Moor was enabled in May after the security review. The values reported for Marlpit are provisional for July and will be confirmed in the first quarterly return. The budget for this activity at Longridge was underspent by 75 per cent because of delays in procurement. The trend at Gallow Hill reversed in August after the works described in the previous section. The standby generator at Marlpit was tested monthly and ran for 8.7 hours during the year. The asset register entry for Quarry Lane was updated in January to reflect the equipment now installed.

### 10.3 Network sensors

Staff from the Central depot supported the work at Fenwick Moor during the busy period in November. Results for Gallow Hill were in line with the previous year, with no change in the operating regime after January. Spare parts for the equipment at Ravensgill are now held at the Riverside depot to shorten repair times. Seasonal variation at Longridge was more pronounced than usual because of the dry spell in late June. A review of the data for Gallow Hill found no evidence of drift in the 76 instruments used for the returns. No adjustment has been made for the transfer of the Upland zone boundary, which affects fewer than 142 properties.

A contractor carried out the work at Gallow Hill under the framework agreement, finishing 3 days ahead of plan. An independent check of the Riverside zone returns in September found 17 minor transcription errors, now corrected. Operational staff at Yarrowby reported that the procedures revised in November were easy to apply on night shifts. A contractor carried out the work at Oakhanger under the framework agreement, finishing 15 days ahead of plan. At Gallow Hill the change compared with the previous year, 4.1 per cent, is within the uncertainty of the measurement.

At Thornby the change compared with the previous year, 5.2 per cent, is within the uncertainty of the measurement. Results for Yarrowby have been restated for January after a meter calibra-

tion error was found. Detail for Netherby and the other sites in the Central zone is held in the operational database. No adjustment has been made for the transfer of the Upland zone boundary, which affects fewer than 202 properties. Performance at Wexcombe was affected for 5 days by a power failure on the distribution network in January. The local authority was informed of the work at Coldharbour in advance, and no objections were received.

Samples from Marlpit were analysed by the accredited laboratory within 9 hours of collection. Operational staff at Kestrel Mere reported that the procedures revised in November were easy to apply on night shifts. A contractor carried out the work at Kestrel Mere under the framework agreement, finishing 17 days ahead of plan. No adjustment has been made for the transfer of the Eastgate zone boundary, which affects fewer than 214 properties. Operational staff at Wexcombe reported that the procedures revised in March were easy to apply on night shifts. Training on the new procedure was given to 2 operators from Thornby and Upwell.

Samples from Hartsfield were analysed by the accredited laboratory within 9 hours of collection. The method used for Hartsfield differs from that used for Pellow Down, so the two values are not directly comparable. The supplier of the equipment at Marlpit confirmed that support will continue for at least 16 years. The long-term trend at Quarry Lane shows a gradual improvement of about 1.9 per cent a year. Staff from the Eastgate depot supported the work at Pellow Down during the busy period in September. Following a near miss at Quarry Lane in May, the access arrangements were changed.

| Month | North Vale | Riverside | Upland | Coastal |
|---|---|---|---|---|
| January | 97.3 | 94.0 | 97.5 | 91.9 |
| February | 94.6 | 97.7 | 95.7 | 98.5 |
| March | 99.6 | 91.5 | 97.2 | 98.4 |
| April | 91.2 | 92.4 | 97.2 | 92.6 |
| May | 94.1 | 95.3 | 95.9 | 94.2 |
| June | 92.0 | 96.6 | 93.3 | 97.6 |
| July | 99.7 | 94.0 | 99.2 | 95.1 |
| August | 96.4 | 98.1 | 99.0 | 93.8 |
| September | 99.2 | 92.5 | 97.1 | 91.8 |
| October | 99.7 | 91.6 | 98.9 | 93.7 |
| November | 91.5 | 99.7 | 95.2 | 91.1 |
| December | 95.1 | 97.2 | 91.4 | 96.3 |

Table 10.1 – Monthly telemetry availability by zone (per cent)

Share of expected readings received by the control system.

Operational staff at Hartsfield reported that the procedures revised in July were easy to apply on night shifts. Performance at Ravensgill was affected for 17 days by a power failure on the distribution network in September. Demand from the Coastal zone was 8.5 per cent higher in July than in the same month a year earlier. Coldharbour and Saltcote together account for 67 per cent of the total for the network. Seasonal variation at Hartsfield was more pronounced than usual because of the dry spell in late June.

The control room logged 183 alarms from Quarry Lane in January, most of them cleared without a site visit. Staff from the Westmoor depot supported the work at Kestrel Mere during the busy period in August. Data from Wexcombe for the first 4 days of May were rejected by the validation rules and infilled. Spare parts for the equipment at Coldharbour are now held at the Coastal depot to shorten repair times. The values reported for Fenwick Moor are provisional for April and will be confirmed in the first quarterly return.

The asset register entry for Hartsfield was updated in January to reflect the equipment now installed. Samples from Thornby were analysed by the accredited laboratory within 12 hours of collection. Demand from the Eastgate zone was 0.3 per cent higher in February than in the

same month a year earlier. Seasonal variation at Ashby Weir was more pronounced than usual because of the dry spell in late July. The local authority was informed of the work at Oakhanger in advance, and no objections were received. Staff from the Riverside depot supported the work at Yarrowby during the busy period in December.

A study of options for Ashby Weir is planned, with a report due to the operating board in October. Samples from Longridge were analysed by the accredited laboratory within 7 hours of collection. Spare parts for the equipment at Marlpit are now held at the North Vale depot to shorten repair times. Operational staff at Quarry Lane reported that the procedures revised in January were easy to apply on night shifts. The control room logged 76 alarms from Netherby in August, most of them cleared without a site visit. The budget for this activity at Gallow Hill was underspent by 19 per cent because of delays in procurement.

## 11 Health, Safety and Environment

The health and safety of staff, contractors and the public, and the protection of the environment, are reported to the operating board every month. This chapter gives the results for the year and the main lessons learned.

### 11.1 Safety performance

There were three reportable injuries during the year, all to contractor staff, and none resulted in absence of more than two weeks. The lost time injury rate fell to 0.21 per 100,000 hours worked.[4]

The asset register entry for Gallow Hill was updated in July to reflect the equipment now installed. The local authority was informed of the work at Yarrowby in advance, and no objections were received. Two inspections at Hartsfield found no defects, and the next inspection is scheduled for December. Staff from the Eastgate depot supported the work at Kestrel Mere during the busy period in December. The site visit to Saltcote in February confirmed that the readings match the local display. Results for Marlpit have been restated for January after a meter calibration error was found.

A review of the data for Pellow Down found no evidence of drift in the 147 instruments used for the returns. The budget for this activity at Hartsfield was underspent by 75 per cent because of delays in procurement. Results for Marlpit have been restated for December after a meter calibration error was found. A study of options for Oakhanger is planned, with a report due to the operating board in January. The method used for Hartsfield differs from that used for Coldharbour, so the two values are not directly comparable.

The supplier of the equipment at Pellow Down confirmed that support will continue for at least 12 years. The planning assumption for Quarry Lane is unchanged at 8.2 per cent above the long-term average. The asset register entry for Wexcombe was updated in January to reflect the equipment now installed. Staff from the Westmoor depot supported the work at Hartsfield during the busy period in May. The local authority was informed of the work at Gallow Hill in advance, and no objections were received.

The budget for this activity at Pellow Down was underspent by 86 per cent because of delays in procurement. The figures for Upwell include an estimate for June, when the meter at the outlet was out of service for 12 days. A contractor carried out the work at Upwell under the framework agreement, finishing 19 days ahead of plan. Detail for Marlpit and the other sites in the Coastal zone is held in the operational database. A study of options for Hartsfield is planned, with a

[4]The injury rate is calculated on hours recorded in the time sheet system and verified by the internal audit team before publication.

report due to the operating board in January. The risk assessment for Gallow Hill was reviewed in April and the control measures were found to be adequate.

Samples from Yarrowby were analysed by the accredited laboratory within 15 hours of collection. Staff from the Westmoor depot supported the work at Coldharbour during the busy period in January. Improvements at Fenwick Moor are expected to save about 14,322 pounds a year in operating costs. The standby generator at Hartsfield was tested monthly and ran for 9.4 hours during the year. Performance at Kestrel Mere was affected for 9 days by a power failure on the distribution network in February.

**11.2 Environmental compliance**

Seasonal variation at Gallow Hill was more pronounced than usual because of the dry spell in late February. Data from Fenwick Moor for the first 5 days of September were rejected by the validation rules and infilled. Remote access to the controls at Oakhanger was enabled in April after the security review. No adjustment has been made for the transfer of the Coastal zone boundary, which affects fewer than 108 properties. Gallow Hill and Pellow Down together account for 93 per cent of the total for the network. Following a near miss at Ravensgill in March, the access arrangements were changed.

The planning assumption for Coldharbour is unchanged at 1.5 per cent above the long-term average. Staff from the Upland depot supported the work at Yarrowby during the busy period in February. Demand from the Coastal zone was 0.9 per cent higher in September than in the same month a year earlier. A study of options for Ashby Weir is planned, with a report due to the operating board in October. The asset register entry for Quarry Lane was updated in June to reflect the equipment now installed. The standby generator at Netherby was tested monthly and ran for 9.5 hours during the year.

Detail for Ravensgill and the other sites in the Upland zone is held in the operational database. The planning assumption for Hartsfield is unchanged at 6.0 per cent above the long-term average. Results for Gallow Hill were in line with the previous year, with no change in the operating regime after March. The trend at Saltcote reversed in February after the works described in the previous section. Staff from the Westmoor depot supported the work at Coldharbour during the busy period in October. Spare parts for the equipment at Upwell are now held at the North Vale depot to shorten repair times.

Improvements at Gallow Hill are expected to save about 39,385 pounds a year in operating costs. A study of options for Kestrel Mere is planned, with a report due to the operating board in January. Results for Oakhanger were in line with the previous year, with no change in the operating regime after January. The trend at Netherby reversed in November after the works described in the previous section. Remote access to the controls at Kestrel Mere was enabled in May after the security review. Demand from the Riverside zone was 9.5 per cent higher in July than in the same month a year earlier.

**11.3 Structural safety of reservoirs**

All service reservoirs and impounding reservoirs were inspected by the supervising engineer as required. No measures in the interests of safety were recommended for the impounding reservoirs, and the statutory reports were filed on time.

Detail for Longridge and the other sites in the Riverside zone is held in the operational database. The asset register entry for Quarry Lane was updated in April to reflect the equipment now installed. Hartsfield and Oakhanger together account for 80 per cent of the total for the network. An independent check of the Central zone returns in July found 3 minor transcription errors, now corrected. The planning assumption for Fenwick Moor is unchanged at 6.0 per cent above

the long-term average. Data from Yarrowby for the first 4 days of December were rejected by the validation rules and infilled.

The control room logged 214 alarms from Gallow Hill in November, most of them cleared without a site visit. Following a near miss at Thornby in August, the access arrangements were changed. A study of options for Fenwick Moor is planned, with a report due to the operating board in January. The values reported for Yarrowby are provisional for October and will be confirmed in the first quarterly return. Demand from the North Vale zone was 3.6 per cent higher in September than in the same month a year earlier.

The site visit to Gallow Hill in January confirmed that the readings match the local display. Marlpit and Coldharbour together account for 45 per cent of the total for the network. The planning assumption for Hartsfield is unchanged at 6.8 per cent above the long-term average. Following a near miss at Oakhanger in November, the access arrangements were changed. The control room logged 186 alarms from Pellow Down in October, most of them cleared without a site visit.

The risk assessment for Quarry Lane was reviewed in March and the control measures were found to be adequate. The method used for Fenwick Moor differs from that used for Marlpit, so the two values are not directly comparable. An independent check of the Coastal zone returns in May found 16 minor transcription errors, now corrected. The asset register entry for Marlpit was updated in December to reflect the equipment now installed. No adjustment has been made for the transfer of the Upland zone boundary, which affects fewer than 115 properties. The standby generator at Kestrel Mere was tested monthly and ran for 6.9 hours during the year.

Remote access to the controls at Yarrowby was enabled in February after the security review. Two inspections at Hartsfield found no defects, and the next inspection is scheduled for December. Night work at Quarry Lane was limited to 13 shifts to reduce noise for neighbouring residents. A study of options for Fenwick Moor is planned, with a report due to the operating board in February. Training on the new procedure was given to 12 operators from Wexcombe and Hartsfield. The figures for Longridge include an estimate for January, when the meter at the outlet was out of service for 16 days.

## 12 Outlook and Investment Programme

The investment programme for the next five years focuses on mains renewal, leakage reduction, resilience of supplies to the coastal zone, and the replacement of ageing telemetry. This chapter summarises the programme and the targets for the coming year.

### 12.1 Investment programme

The programme for the coming year includes the renewal of 48 kilometres of mains, a second transfer main to the coastal zone, and the refurbishment of the filters at two treatment works. **Capital expenditure remains subject to board approval** of the detailed business cases, which are expected in the spring.

The telemetry upgrade at the outstations will continue with the replacement of the remaining radio links by fibre or cellular connections. The work is planned zone by zone, starting with the Upland zone, where radio coverage is poorest and data gaps are most frequent.

A review of the data for Longridge found no evidence of drift in the 197 instruments used for the returns. The figures for Fenwick Moor include an estimate for June, when the meter at the outlet was out of service for 6 days. The local authority was informed of the work at Yarrowby in advance, and no objections were received. The risk assessment for Ashby Weir was reviewed

in December and the control measures were found to be adequate. Operational staff at Saltcote reported that the procedures revised in April were easy to apply on night shifts. Remote access to the controls at Hartsfield was enabled in November after the security review.

The risk assessment for Ashby Weir was reviewed in January and the control measures were found to be adequate. Improvements at Fenwick Moor are expected to save about 43,353 pounds a year in operating costs. The figures for Ashby Weir include an estimate for March, when the meter at the outlet was out of service for 11 days. The values reported for Marlpit are provisional for February and will be confirmed in the first quarterly return. No adjustment has been made for the transfer of the Eastgate zone boundary, which affects fewer than 97 properties. Samples from Gallow Hill were analysed by the accredited laboratory within 10 hours of collection.

Samples from Netherby were analysed by the accredited laboratory within 3 hours of collection. The supplier of the equipment at Thornby confirmed that support will continue for at least 16 years. Detail for Thornby and the other sites in the Upland zone is held in the operational database. The figures for Yarrowby include an estimate for May, when the meter at the outlet was out of service for 18 days. The values reported for Pellow Down are provisional for April and will be confirmed in the first quarterly return.

Following a near miss at Upwell in December, the access arrangements were changed. Operational staff at Upwell reported that the procedures revised in September were easy to apply on night shifts. The figures for Ravensgill include an estimate for November, when the meter at the outlet was out of service for 5 days. Improvements at Fenwick Moor are expected to save about 25,815 pounds a year in operating costs. At Kestrel Mere the change compared with the previous year, 9.0 per cent, is within the uncertainty of the measurement.

### 12.2 Targets for the coming year

Table 12.1 summarises the main performance indicators for 2025 and the targets for the coming year. Negative changes indicate an improvement for leakage and interruptions.

| Indicator | 2025 | Target | Change (%) |
|---|---|---|---|
| Leakage (megalitres per day) | 21.4 | 20.8 | −2.6 |
| Mains bursts per 1,000 km | 118 | 112 | −5.1 |
| Supply interruptions (minutes per property) | 6.2 | 5.5 | −11.3 |
| Water quality compliance (per cent) | 99.96 | 99.97 | +0.01 |
| Energy intensity (kWh per cubic metre) | 0.47 | 0.46 | −2.1 |

Table 12.1 – Summary of performance indicators

A study of options for Marlpit is planned, with a report due to the operating board in May. The values reported for Coldharbour are provisional for May and will be confirmed in the first quarterly return. An independent check of the Westmoor zone returns in April found 9 minor transcription errors, now corrected. Training on the new procedure was given to 7 operators from Thornby and Marlpit. Results for Quarry Lane have been restated for August after a meter calibration error was found. Samples from Marlpit were analysed by the accredited laboratory within 17 hours of collection.

The control room logged 87 alarms from Oakhanger in April, most of them cleared without a site visit. The budget for this activity at Yarrowby was underspent by 68 per cent because of delays in procurement. Results for Pellow Down have been restated for October after a meter calibration error was found. A study of options for Saltcote is planned, with a report due to the operating board in May. Staff from the Coastal depot supported the work at Yarrowby during the busy period in March.

Seasonal variation at Pellow Down was more pronounced than usual because of the dry spell in late December. Performance at Fenwick Moor was affected for 18 days by a power failure on

the distribution network in June. Night work at Fenwick Moor was limited to 14 shifts to reduce noise for neighbouring residents. The method used for Fenwick Moor differs from that used for Yarrowby, so the two values are not directly comparable. The standby generator at Pellow Down was tested monthly and ran for 6.2 hours during the year.

The value for Fenwick Moor is the highest in the Westmoor zone, reflecting the size of the area it serves. The budget for this activity at Gallow Hill was underspent by 77 per cent because of delays in procurement. An independent check of the Westmoor zone returns in February found 19 minor transcription errors, now corrected. The budget for this activity at Ravensgill was underspent by 92 per cent because of delays in procurement. Night work at Quarry Lane was limited to 5 shifts to reduce noise for neighbouring residents. Results for Wexcombe have been restated for August after a meter calibration error was found.

### 12.3 Risks

The main risks to delivery are the availability of specialist contractors for trunk main work, the lead time for electrical equipment, and the weather in the summer peak. Each risk has an owner and a mitigation plan, which are reviewed by the programme board every quarter.